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“Measuring energy consumption of online ads on social networks”

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ABSTRACT

In the last years, there has been an increasing interest in understanding the consumption of energy and emissions of internet and technological processes. In this way, measuring the energy consumption of posts and ads on social media becomes an important point of research, and because it is currently an unknown subject, it is of immense importance to study and report it. To do so, this work presents an automated methodology that collects posts and ads from social networks (Facebook and Twitter), isolates them and analyses the energy consumption associated with their rendering process.

In order to measure the energy consumption of ads and posts, it is necessary to collect a vast amount of data. In this work a dataset of 508 ads and 6,769 posts, including 1,598 entries for Facebook and 5,679 entries for Twitter, was collected from these social media platforms, together with different parameters for each of them, such as the amount of likes or shares.

The data collection process is one of the main points of this work, and the scraping methodology developed to retrieve the data was the most time-consuming phase. It involves identifying the specific elements of ads and posts on social media and isolating their HTML content by using a sophisticated python program.

The other main subject of this work involves the measurement and results of the energy consumption of ads and posts on Facebook and Twitter. To do so, an accepted scientific methodology of energy measurement, referenced in the literature, was used. This methodology loads the element in an isolated environment to determine the energy consumption of this element independently of the baselined consumption of the device.

Using this approach, the results of this work demonstrate that ads consume in median around 1.6 times more energy than posts on Facebook. Moreover, it is worth mentioning that Facebook ads consume about 8.7 times more energy than on Twitter.

This proves that the energy consumption of ads and posts on social media is a factor to consider when the energetic cost in society is discussed. Works like the following one represent an important step in understanding the emissions of social networks to adopt the relevant measures, both at a social and regulatory level.

Keywords: Facebook, Twitter, advertising, software, embeddings

DEDICATION

I would like to express my gratitude to all those who have accompanied me and supported me throughout the duration of my bachelor's degree, particularly my parents Gloria Macías and Felipe Castellano, who have believed in me at every step of this journey, and my brother Felipe, who served as an inspiration to me.

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ACRONYMS

API	<i>Application Programming Interface</i>
CO2	<i>Carbon diOxide</i>
CPA	<i>Cost Per Action</i>
CPC	<i>Cost Per Click</i>
CPM	<i>Cost Per Mille / per thousand impressions</i>
CPU	<i>Central Processing Unit</i>
CSV	<i>Comma-Separated Values</i>
EIA	<i>Energy Information Administration</i>
EU	<i>European Union</i>
GB	<i>GigaByte</i>
gEqCO2	<i>gram Equivalent of Carbon dioxide</i>
GHz	<i>GigaHertz</i>
GIF	<i>Graphic Interchange Format</i>
GPU	<i>Graphics Processing Unit</i>
HTML	<i>HyperText Markup Language</i>
ID	<i>IDentifier</i>
kgEqCO2	<i>kilogram Equivalent of Carbon diOxide</i>
kWh	<i>kiloWatt per hour</i>
mAh	<i>milliAmpere per hour</i>
OS	<i>Operating System</i>
RAM	<i>Random Access Memory</i>
UN	<i>United Nations</i>
URI	<i>Uniform Resource Identifier</i>

URL	<i>Uniform Resource Locator</i>
USD	<i>United States Dollars</i>
UUID	<i>Unique IDentifier</i>
VSC	<i>Visual Studio Code</i>

1. INTRODUCTION AND MOTIVATION

In this first section, the topic and motivation of this work are introduced to put into perspective its importance and the objectives of this study are presented to define what is expected to be achieved and why it is extremely relevant for society. An unprecedented set of insights are then listed to encapsulate the main results of this work.

1.1 INCREASING USAGE OF SOCIAL MEDIA PLATFORMS

During recent years, there has been a rise on the use of social media platforms in today's society, as well as an exponential increase in the amount of social media users around the world. There are currently around 5 billion social media identities across different platforms, accounting for approximately 62% of the total global population, with a growth rate of more than 8 new users per second [1]. As a result, the whole world spends about 12 billion hours on social media every day or, in other words, 1.35 million years worth of time [1].

Social media consists of scrolling down a feed that is constantly loading ads and posts for the user to visualize. For example, in June 2021, the average Facebook user liked a median of 11 posts, left a median of 5 comments, reposted a median of 1 post, and clicked on a median of 12 ads, during the previous 30 days [2]. Taking this into consideration, social media advertising can be considered the most powerful tool to reach all kinds of people and to economically finance social networks, which presents a huge opportunity to companies that are looking to promote their products or services in a very personalized manner.

Therefore, all different types of companies are now recurring to social media to reach their target audience by using online advertising on these platforms. In 2021, the spending on social media advertisements was estimated at 116 billion USD for that year, with a prediction of an appalling cost of 262 billion USD throughout the year 2028 [3].

This means that social media networks are making very large amounts of money with their online advertising tools, which is their most important source of revenue [4]. They earn money every time a user sees an online ad or when they click on it. As a consequence, social media networks are very interested in increasing their amount of users, but most importantly, in making sure they are actively using their platforms.

Figure 1 represents the ranking of 10 of the most influential social media networks in terms of advertising reach. Facebook reached around 2.24 billion users through their online ads and had a total of 2.989 billion monthly active users by April 2023 [2]. On the other hand, Twitter (X) reported 368 million active users in April 2024 [5], which still poses a potentially large advertising audience.

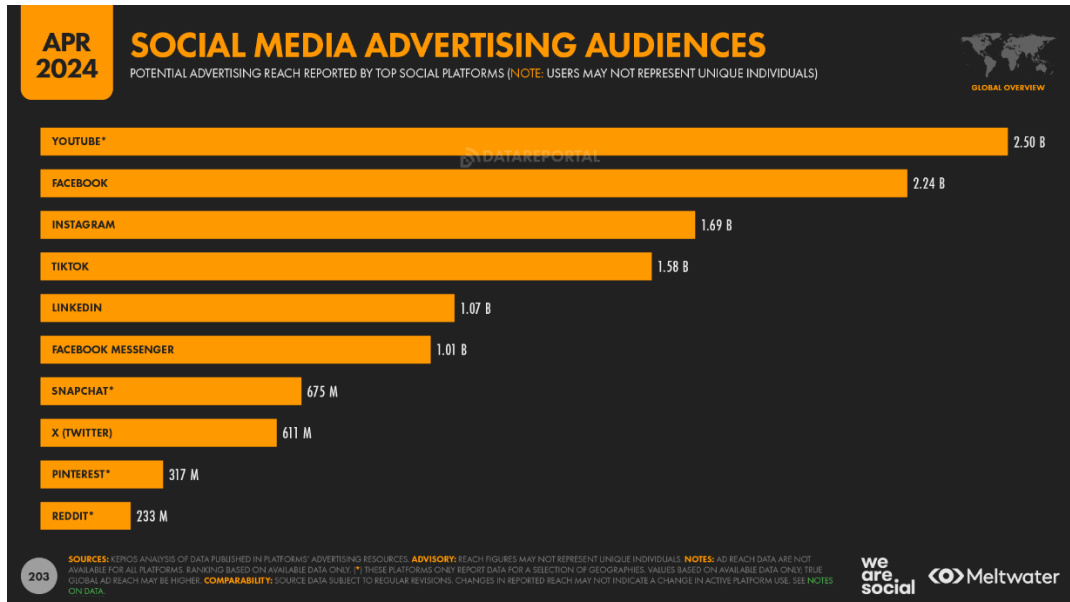


Figure 1 – Social media potential advertising audiences. Source: DataReportal

1.2 MOTIVATION: THE ENERGY CONSUMPTION OF SOCIAL MEDIA

However, the data previously presented raises a series of ethical concerns on both the social media platforms and the online advertising companies. The outrageous amount of users across these platforms means that a very large amount of energy is being consumed due to the display of online ads and posts on their electronic devices. This fact questions the environmental impact that digital advertising has and sets a new ground in order to explore the extent of it.

The amount of emissions and energy attributed to social networks is an important point of discussion. Unveiling this will benefit society to take greener steps into the usage of social networks. According to a previous projection by Greenspector [6], taking into account the average carbon impact (1.15 gEqCO₂ per minute) of the 10 social media platforms in France shown in Figure 2, every user is approximately consuming 165.6

gEqCO₂ per day. Scaling this amount, each user would be consuming about 60 kgEqCO₂ per year, or the equivalent of traveling 535 km in an average light vehicle.

Figure 2 also depicts more interesting insights that serve as motivation to this work. Out of the 10 social media networks, TikTok is the undisputed leader in higher energy consumption per hour with 15.81 mAh, followed by Facebook with 12.36 mAh. The last position is claimed by YouTube with 8.58 mAh, almost half of TikTok's rate.

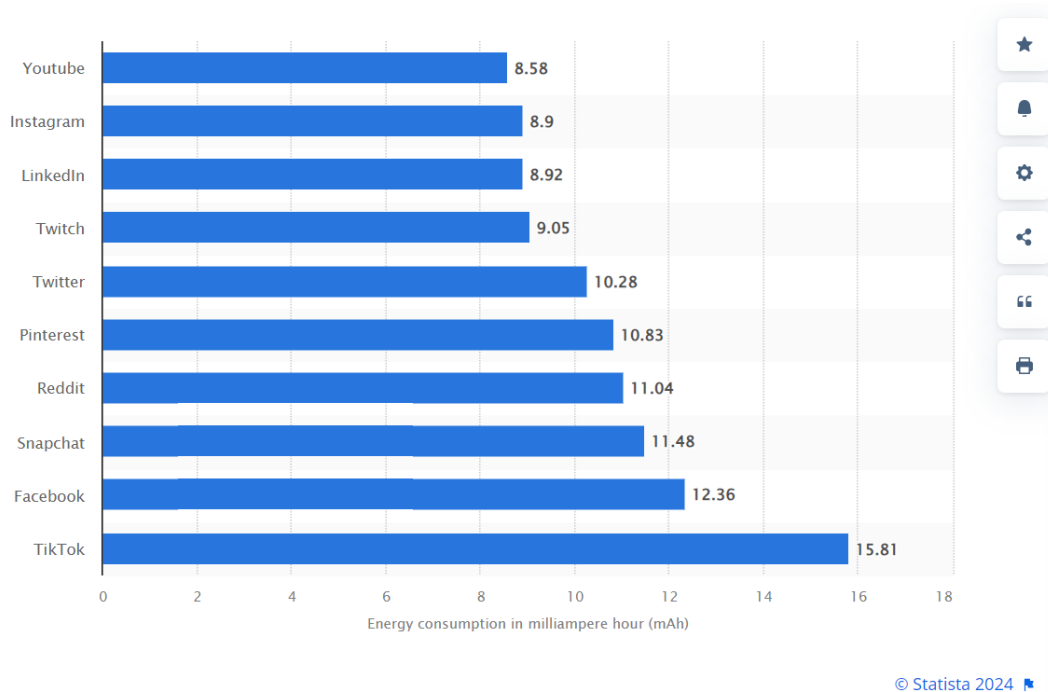


Figure 2 – Energy consumption rate per hour of top social networks. Source: Statista

1.3 OBJECTIVES OF THIS WORK

As we have seen, measuring the emissions and energy consumption of the information presented on social networks is a very relevant topic of research. This work intends to be a clear step forward into understanding the energy consumption on two of the most important social networks of all time: Facebook and Twitter.

We focus on Facebook and Twitter for the following reasons:

- They are the top 1 and top 12 social networks in terms of amount of users, respectively [7].
- They have been high up in this ranking for many years, so they are consolidated social networks. This is relevant because advertisers can easily

create ads and this work is intended for platforms where ads (and posts) can be measured.

- They internally upload their content onto their platforms, so there are no third parties involved during this process.
- This work is interested in measuring the consumption of the rendering process of ads and these platforms are relevant because they have different types of content, not only video content, such as TikTok or YouTube.

In order to make use of the huge potential that these social networks present, the main objectives of this work are defined as the following:

1. Develop an automatic methodology that allows the extraction and embedding of ads and posts to measure their energy consumption.
2. Determine the factors that influence the amount of energy consumed per ad and posts and obtain results on the energy differences between ads and posts within Facebook and Twitter and between them.

1.4 MAIN INSIGHTS

The main insights unveiled throughout the course of this work present new and interesting discoveries on the energy consumption and consist of the following:

1. This project reveals an unprecedented study on measuring energy consumption of ads and posts on social networks. For that, it relies on a total of 508 ads and 6,769 posts on Facebook and Twitter.
2. Facebook ads consume 1.6 times more energy than posts do and 1.75 times more energy on Twitter
3. Facebook consumes around 8.3 times more energy than Twitter in posts and 8.7 times more in ads. Facebook also consumes approximately 1.2 times more energy than Twitter in ads and 1.4 times more in posts per millisecond of loading time.
4. If the energy consumption and loading time of Facebook ads were to be reduced by 1%, the total yearly energetic cost for the end users would be decreased by 174,928€.

2. STATE OF THE ART

In this section, ads and posts on social networks are discussed as it is very important to understand their structure to later define the extraction methodology. In section 2.1, online advertising is explained, and the visual elements of ads are identified and differentiated from posts.

Section 2.2 then reveals the hidden elements of ads and posts that are relevant to this work because they are responsible of how their content is loaded into the page.

Sections 2.3 and 2.4 focus on advertising on Facebook and Twitter, respectively, which is important to this work as it presents several research branches explored in this study.

2.1 ONLINE ADVERTISING

Online advertising, or digital advertising, is the most common marketing technique used by social media networks, such as Facebook or Twitter. One of the most obvious forms of online advertising on social networks is known as “News Feed Ads”, which is a type of web banner advertising that consists of showing graphical ads to their users [8]. Some examples of this advertising form are Facebook’s “Sponsored Stories” or Twitter’s “Promoted Tweets”.

2.1.1 Understanding Online Advertising

The goal of online advertising for social media networks is generating as much revenue as possible. The parties involved in this revenue system are the following:

1. Social networks: they serve as intermediaries between companies and users. They provide an advertising platform with a specific audience that allows companies to design sophisticated advertising techniques that maximize the number of users reached through their ads.
2. Advertising companies: their objective is to promote a product, service, or idea by targeting a specific audience in the social media platform. They are the parties that spend money to make use of the platform space where they want to publish their ads.

3. Social media users: correspond to the social media identities that are targeted by the advertising companies and interact with the different ads across the social network.

2.1.2 Visual Appearance of Ads

Ads are typically displayed on the user's feed so that they are frequently seen while scrolling the page. They usually have a similar appearance to other elements of the feed, such as posts, allowing the ad to blend in with the user's feed and making it more appealing to the user.

Social media ads can contain different elements, such as images, videos, scrollable items, audio, and animations, which may be indistinguishable from other posts on a user's feed that contain similar items.

For instance, let's compare the appearance of an ad and a post on Facebook's feed, shown in Illustration 1, to identify the elements that differentiate them.

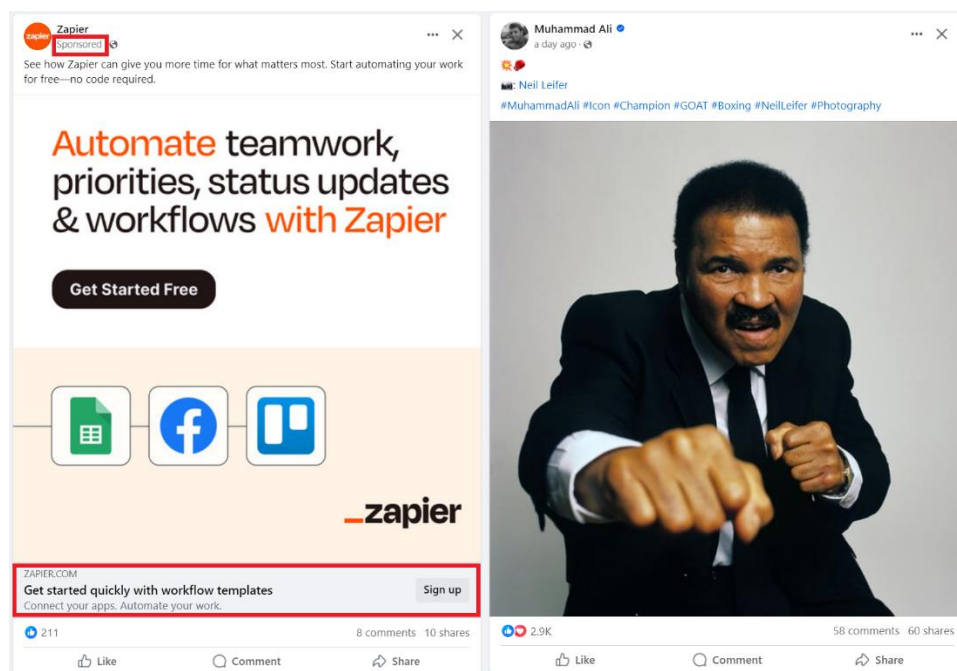


Illustration 1 – Visual difference between ads (left) and posts (right) on Facebook

As it can be seen from Illustration 1, the ad on the left side is very similar to the post on the right side, with only a few visual differences that the user may be able to identify, highlighted in red.

The main difference is that the ad contains the word “Sponsored” below the name of the advertising account, while the post does not. This word is shown in a very small font so that the user’s focus is on the content of the ad itself and not on the fact that it is an ad.

Also, most ads contain a banner below its graphic content that encourages the user to click on it to access the advertising product or service, with the intention of buying it or subscribing to it, while posts do not have this element. This banner has a noticeable size so that the users are attracted and visit the advertiser’s page if they are interested in the ad.

Both ads and posts give the user an option to interact with it by viewing its content, liking it, commenting on it, and sharing it with others, so the user can barely feel any difference on their feed while scrolling down.

2.2 ADVERTISING ON FACEBOOK

In this subsection, advertising on Facebook is discussed to understand which ads are relevant to this work.

2.2.1 Types of Ads on Facebook

On Facebook’s web, 2 different types of ads can be distinguished based on their location on the page:

1. Right-side banner ads: they are located on the right side of Facebook’s home page, under the title “Sponsored” and show 2 static ads, each containing an image and a phrase. They are “traditional ads”, as they are loaded independently and not along with posts. These ads will not be studied in this work.

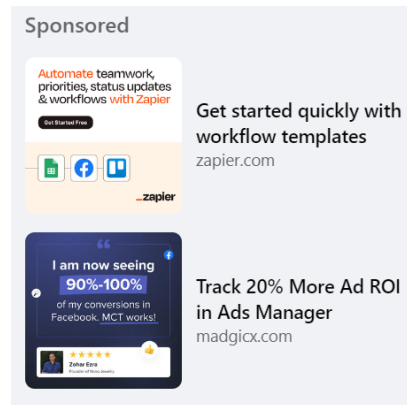


Illustration 2 – Right-side banner ads

2. **News Feed Ads:** they can be found while scrolling down on Facebook's feed and have a similar aspect to posts. They can have different visual elements, including images, videos, and scrollable images. This work will focus on studying this type of ad as they are the ones that are loaded together with the rest of the content in terms of internal calls and have to be differentiated from posts to make a fair comparison.

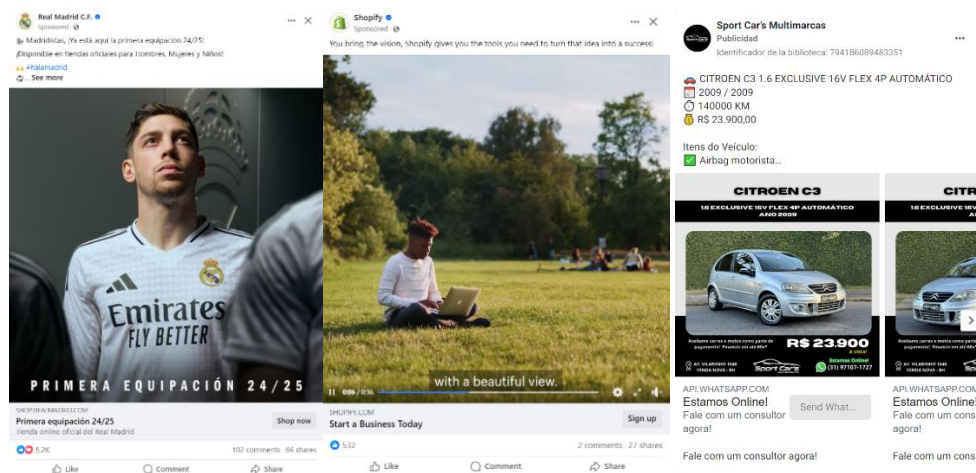


Illustration 3 – Ads with images, videos, and scrollable images (left to right)

2.2.2 Ad Library

Meta, the company that owns Facebook, has provided an open-source ad library for everyone to use [9]. This library allows every user, without any authentication required, to search for every ad that has been published on Facebook. This search can be narrowed down by defining the following parameters:

- Country where the ad is being displayed
- Category of ads, such as social or political aspects
- Keywords or name of the advertising party

Below, Illustration 4 depicts the search tool provided by the ad library to make a personalized search of ads On Facebook.

The screenshot shows the 'Search ads' section of the Facebook Ad Library. It includes a header with the title 'Search ads' and a sub-header 'Set your location and choose an ad category to start your search. View search tips.' Below this, there are two dropdown menus: 'Spain' and 'Ad category'. To the right of these is a search bar with a magnifying glass icon and the placeholder text 'Choose an ad category'. At the bottom left, there is a button labeled 'Saved searches' with a bookmark icon.

Illustration 4 – Search tool on Facebook Ad Library

As it can be seen from Illustration 5, a search was conducted that chose Spain as location of the ad, all ad categories, and used the keyword “tennis”. The results of that personalized search can be seen below:

The screenshot displays four search results for the keyword 'tennis'. Each result card includes the following information:

- Library ID:** 1827333924408648, 1247998176165141, 806842554310870, and 803650384636764.
- Status:** Active for the first three, and Inactive for the last one.
- Start Date:** Started running on May 28, 2024, May 9, 2024, and May 8, 2024.
- Platforms:** Facebook, Instagram, and Messenger.
- Ad Details:**
 - Ad 1: 'Las últimas colecciones de Adidas, Nike, Wilson dedicadas a Roland Garros.' with a 'Watch More' button.
 - Ad 2: 'Las últimas colecciones de Adidas, Nike, Wilson dedicadas a Roland Garros.' with a 'Watch More' button.
 - Ad 3: 'Los best sellers 2023 en la Tennis Zone. Compruébelo ahora en www.tennis-zone.es' with a 'Shop Now' button.
 - Ad 4: 'Descubre nuestra última colección de ropa de tenis. Prepárate para conquistar la pista.' with a 'Shop Now' button.

Illustration 5 – Results from search tool on Facebook Ad Library

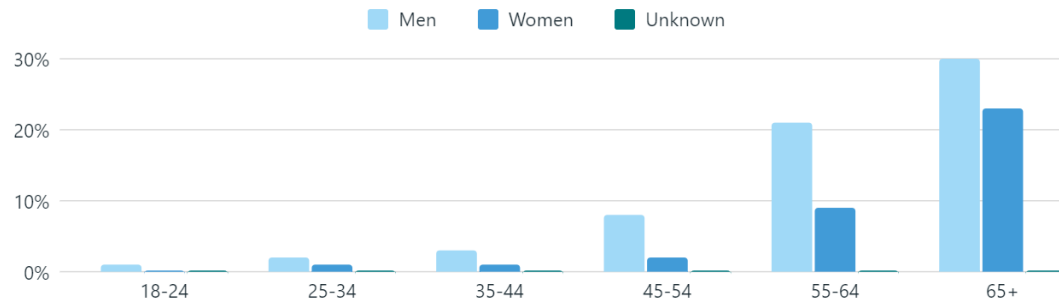
The search returned the ads that met the search filters and were already launched on Meta’s platforms, starting with the most recent ones. The results show relevant ads related to tennis in Spain.

In addition, there is some interesting information provided for the ads displayed by conducting a search:

- Unique ID, corresponding to a large number
- Status, active or not
- Launch date of the ad
- Platforms where the ad is being shown, which can be Facebook and Instagram
- Categories of the ad
- Estimated audience to be reached by the ad, based on the number of accounts that match the requisites to be included in the target audience of the advertisers on the day of creation of the ad (political ads only)
- Amount of money spent by the advertisers during the ad's lifespan (political ads only)
- Number of views the ad has obtained, including multiple views from single users (political ads only)

Furthermore, detailed information about some ads is also disclosed when clicking on them (usually for politically-related ads), including contact information of the advertising party, a graph showing the number of viewers per age range and gender, the specific location of the viewers (e.g. autonomous community within Spain), and the total amount of money spent by the advertisers since they first started creating ads in Meta and since the previous week.

Illustration 6 shows the age and gender distribution of viewers and the amount of money spent in total and in the previous week for a politically-related ad intended for citizens of the Community of Madrid.

Age and gender ⓘ**Location** ⓘ

Comunidad de Madrid

**Total amount spent €49,031**

Apr 15, 2019 - Jun 7, 2024 in Spain

The estimated total amount of money this advertiser has spent on ads about social issues, elections or politics.

Disclaimers**Spend per disclaimer**

Maria Revilla Hernando

€293

PSOE Madrid

€48,738

Last week's spend €3,372

7 days • Jun 1 - Jun 7, 2024 in Spain

The estimated amount of money this advertiser spent during the last week on ads about social issues, elections or politics.

Disclaimers**Spend per disclaimer**

PSOE Madrid

€3,372

Illustration 6 – Disclosed information by Meta on social/political ads

2.3 ADVERTISING ON TWITTER

In this subsection, advertising on Twitter is discussed to understand which ads are relevant to this work.

2.3.1 Types of Ads on Twitter

Elon Musk's social network, Twitter (recently renamed to X), only displays ads to their users on their home page, while scrolling down. These news feed ads are very similar to

other posts and can contain images or videos, while also allowing users to interact with the ad itself as they would do with a normal post, by viewing, liking, commenting, or reposting.

Two different ads on Twitter, containing different types of content, images and videos, are depicted below in Illustration 7.



Illustration 7 – Image (left) and video (right) ads on Twitter

2.3.2 Ads Repository

Twitter provides a tool called Ads Repository [10], which for the main goal of this project, is not very useful because it does not provide any relevant information to identify elements in ads (or posts). This tool allows everyone to search for a specific advertiser, country, and date range, and then a report is provided in a CSV file, which includes information about all the ads launched that meet the search parameters.

The information given in the file includes the advertiser, the start and end dates, the targeted segments, the number of impressions and users reached, and, if any, the circumstances of removal of the ads.

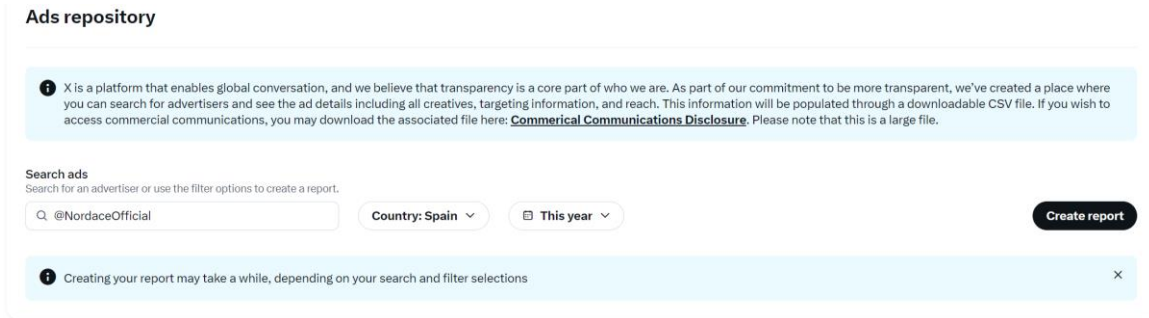


Illustration 8 – Twitter’s Ads Repository search tool

2.4 THE HIDDEN PART OF ADS AND POSTS

After realizing the visual differences between ads and posts, it is also important to understand the hidden elements in them. These hidden elements correspond to a series of resources that act on the background as calls and are the ones that allow the ad or post to be loaded onto the page. These calls are the ones that generate the consumption of energy, so it is important to understand where they come from.

Illustration 9 depicts the internal calls on social networks that can be found in the “Network” tab of the inspection page.

Name	Status	Protocol	Scheme	Domain	Type	Initiator	Size	Time
445003848_802768158623777_7...	200	h3	https	scontent-mad2...	jpeg	8EWdDKKq	(disk cache)	1 ms
305576871_463965995782215_8...	200	h3	https	scontent-mad2...	jpeg	8EWdDKKq	(disk cache)	2 ms
447961394_848893837289427_2...	200	h3	https	scontent-mad2...	jpeg	8EWdDKKq	(disk cache)	1 ms
bulk-route-definitions/	200	h3	https	www.facebook...	xhr	8EWdDKKq	7.7 kB	178 ms
graphql/	200	h3	https	www.facebook...	xhr	8EWdDKKq	88.0 kB	897 ms
4b1hXcZV1Xjs?_nc_x=b_v6UNh...	200	h3	https	static.xx.fbcdn...	script	uhj1dVL_nN	3.4 kB	195 ms
dataapplication/cr...	200	data	data		script	uhj1dVL_nN	(memory ca...	1 ms
dataapplication/cr...	200	data	data		script	uhj1dVL_nN	(memory ca...	0 ms
412291349_225326713942549_2...	200	h3	https	scontent-mad2...	jpeg	8EWdDKKq	1.3 kB	117 ms
447791613_331694103305801_2...	200	h3	https	scontent-mad2...	jpeg	8EWdDKKq	60.3 kB	231 ms
398010978_238159625936745_6...	200	h3	https	scontent-mad2...	jpeg	8EWdDKKq	1.7 kB	119 ms
448132605_378052301947476_1...	200	h3	https	scontent-mad2...	jpeg	8EWdDKKq	84.7 kB	354 ms
326264476_1204763143808860...	200	h3	https	scontent-mad2...	jpeg	8EWdDKKq	1.4 kB	105 ms
447689359_982295770238005_2...	200	h3	https	scontent-mad2...	jpeg	8EWdDKKq	65.1 kB	168 ms
244034637_469659603726782_...	200	h3	https	scontent-mad2...	png	8EWdDKKq	2.0 kB	104 ms
448092138_1003365084489220_...	200	h3	https	scontent-mad2...	jpeg	8EWdDKKq	47.0 kB	159 ms
448148249_1003365121155883_...	200	h3	https	scontent-mad1...	jpeg	8EWdDKKq	48.4 kB	161 ms
448089930_1003365111155884_...	200	h3	https	scontent-mad2...	jpeg	8EWdDKKq	41.1 kB	153 ms
448026602_1003365131155882_...	200	h3	https	scontent-mad2...	jpeg	8EWdDKKq	38.6 kB	149 ms
1f1fa_1f1f8.png	200	h3	https	static.xx.fbcdn...	png	8EWdDKKq	651 B	73 ms

Illustration 9 – Internal calls of ads and posts on Facebook

3. RELATED WORK

In this section, similar works to this one will be explored, highlighting the main differences and similarities between them.

For Facebook, there is a project that developed a scraper for this social network and launched an open-source python library [11], to be used without the need of an API key and on Facebook public pages. This scraper can return information about posts, profiles, or groups, in the form of dictionaries. Similarly to this work, it can also retrieve interaction parameters, such as the number of likes in a post, and some URLs and texts in the post.

However, this library is not able to collect the entire HTML content of ads and posts as it is done with the scraping software developed in this work, as well as successfully identifying and differentiating ads and posts.

Similarly, there is another open-source project in GitHub that designed a Twitter scraper [12]. This is also available as a python library and can retrieve data about tweets, such as the number of replies or the text of the tweet. But again, it is not able to obtain the embedding content of ads and posts on Twitter as this work can do.

Moreover, the other main difference between this project and the open-source scrapers mentioned above is that this work takes the data collected from both social networks and measures the energy consumption of these ads and posts to discover hidden trends and make unprecedented comparisons.

Nevertheless, there are two different related works that are focused on measuring the energy consumption of online advertising.

The first work, CarbonTag [13], focuses on a browser-based method for approximating the energy consumption of online ads by investigating their rendering process, but just for ads on the web and not on social networks, as this work does. Still, their measurement methodology is useful for this work to obtain results on the energetic cost of ads on social media.

The second work, Scope3 [14], measures media and advertising emissions on the web, but again, does not measure energy consumption on social media.

4. METHODOLOGICAL FRAMEWORK

The next section encompasses the bulk of this project and details a complex and automated methodology to scrape ads and posts from Facebook and Twitter. It defines a pioneer method of identifying ads and posts from these social networks and sets new ground for the research of energy consumption on social media.

Section 4.1 describes the main objectives and challenges that are present in the development of this innovative methodology.

Sections 4.2 and 4.3 explain the data collection process of ads and posts on Facebook and Twitter, respectively. They start with a research phase to explore the different options to identify elements on social networks, followed by the detailed description of the automated software developed for each of the platforms to collect ads and posts along with their parameters (and including relevant pieces of code), and ending with the embedding methodology designed to obtain HTML files with the content of ads and posts.

The last part in this block, section 4.4, defines the widely accepted scientific methodology used to measure the energy consumption of HTML files, in this case holding the content of ads and posts.

4.1 OBJECTIVES AND CHALLENGES

As stated in section 1.3, the objective of this work is to measure the amount of energy consumed by ads and posts on social networks, specifically on Facebook and Twitter. To accomplish this, a database of ads and posts needs to be collected in order to conduct energy measurements on them. However, this poses a series of challenges that need to be solved in order to obtain a relevant dataset.

Usually, ads that are displayed on the web can be easily identified on the page's HTML code as a separate iframe, which is an HTML element that loads another HTML page from within the webpage [15]. An example of the iframe content of an ad on the web is shown below in Illustration 10, highlighted in red.

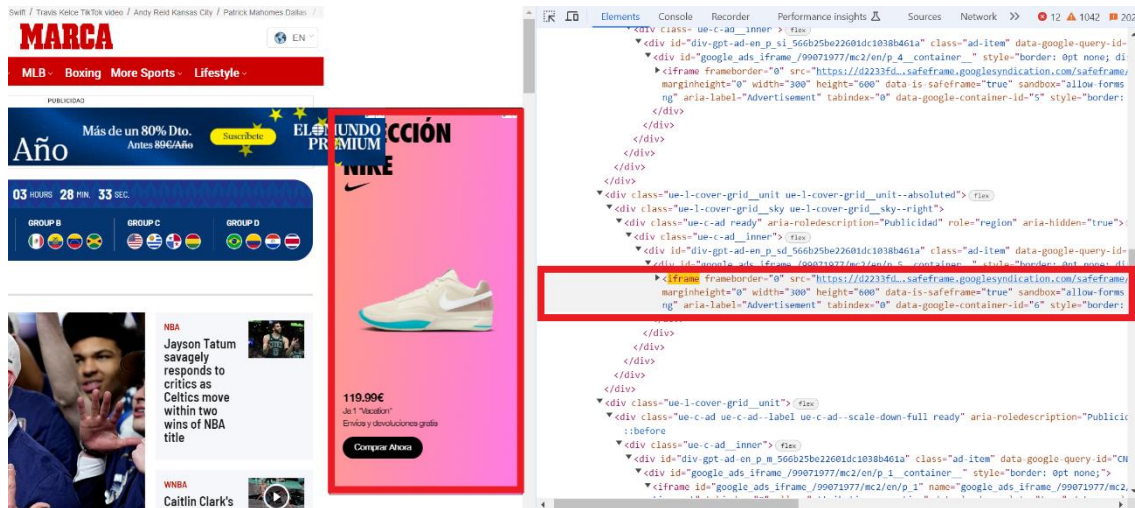


Illustration 10 – *Iframe element of an ad, highlighted in red, on Marca's website*

On the other hand, ads on social networks do not appear as an `iframe` on the web, instead they appear as another element of the webpage that is hard to differentiate from other elements. Illustration 11 depicts the “`div`” element that encloses the content of an ad on Facebook, which is practically indistinguishable from any other element on the page.

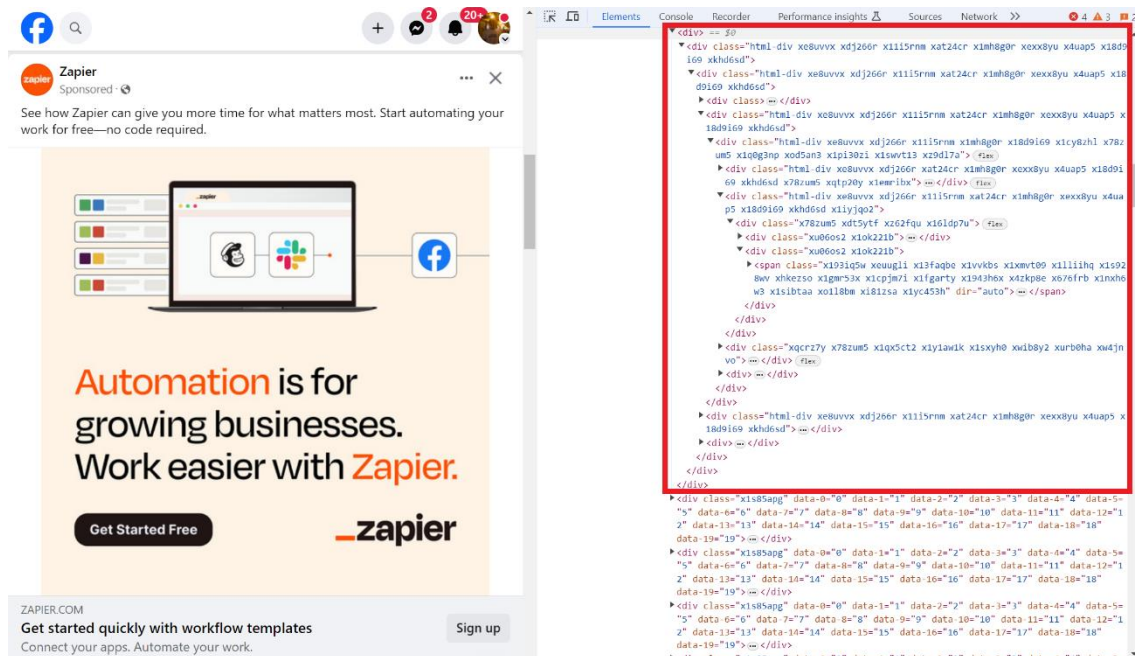


Illustration 11 – *HTML content of an ad on social networks (Facebook)*

As a result, obtaining a dataset of ads from these social networks is not an easy task, as every element of the ad needs to be correctly identified and not mistaken with some other element on the page. For example, in order to collect ads from the news feed, these ads need to be differentiated from the rest of the posts that may appear too, even after scrolling down.

In addition, the iframe from ads on the web is easily obtained from the HTML content, but for social networks it is not possible to just get the “div” element of an ad because the rest of the elements that load the ad would be lost. Therefore, this work also finds a method to obtain the full embedding content, including these background elements and calls.

4.2 DATA COLLECTION ON FACEBOOK

This section proceeds with the methodology to collect ads and posts on Facebook and to obtain their embeddings.

4.2.1 Research Phase

In order to find the most adequate way of collecting ads on Facebook, a research phase was conducted to explore the different possibilities of identifying ads, scraping Facebook’s page and optimizing the process.

Visually, as previously stated in 2.1.2, ads differ from posts by containing the word “Sponsored” and a banner to click on. By exploring the HTML code of these elements in the page, it was found that the word contains an href element corresponding to the URI “<https://www.facebook.com/ads/about>” followed by a combination of characters. Additionally, the banner was found to contain an xlink that redirects the user to the advertiser’s external page.

To find the most adequate method of identifying ads and posts separately on Facebook, 3 different options were explored in this work:

- The HTML code of Facebook’s page was consulted by right-clicking on the desired spot on the page, and then clicking on the bottom option “Inspect”. This opens a window on the right side of the page (using Google Chrome), that shows multiple information about the elements and activities in the

browser. Illustration 12 shows the elements menu that contains the HTML content of the page, where the href element of an ad is highlighted.

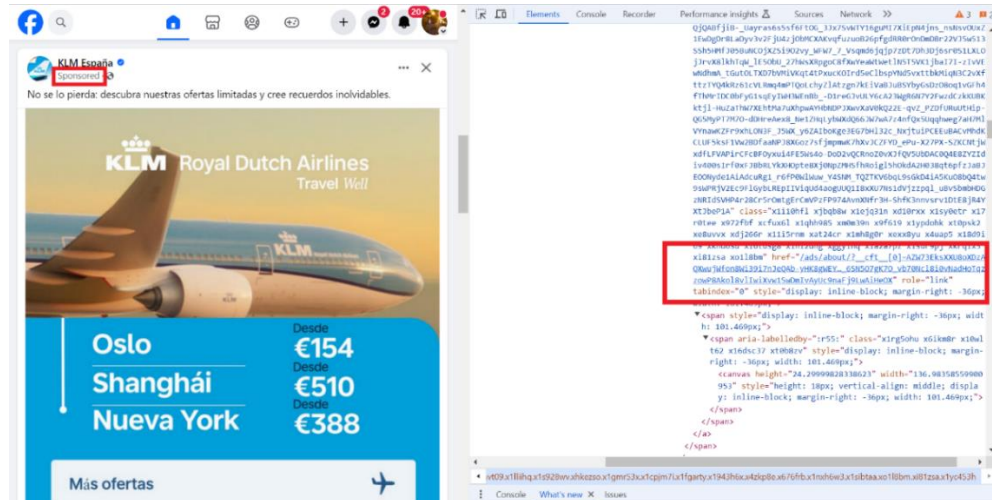


Illustration 12 – Inspection of HTML code on Facebook's ads

- Facebook's Ad Library was explored to find a way to directly collect ads from its page, as the potential number of ads to be obtained would be huge. As aforementioned in section 2.2.2, every ad contains a unique ID in the ads library, and if that ID were to be found on Facebook's feed, ads could be identified very easily. However, this ID was nowhere to be found on Facebook's feed, so it was of no use for this work.
- Ad blockers were investigated, as they successfully remove ads from Facebook's page, which means they are able to differentiate them, but again it was found to be a dead end for the task of obtaining a database of ads.

However, all of these options were not successfully developed. Even though the href element "<https://www.facebook.com/ads/about/>", which is only found in the HTML content of ads and not for posts, was explored and tested programmatically, it did not give very decent results in terms of identifying all ads.

Nevertheless, by investigating Facebook's feed, an embedding option provided by Facebook itself was found, which was one of the main objectives of this phase, by clicking on the 3 circles button on the top right corner of ads. Then, a set of options was shown and by clicking the option "Embed", Facebook provides the corresponding iframe code with an option to copy it, as well as the preview of the embedding of the ad.

Illustration 13 represents the 3-step process needed to obtain the HTML code to embed an ad or post.

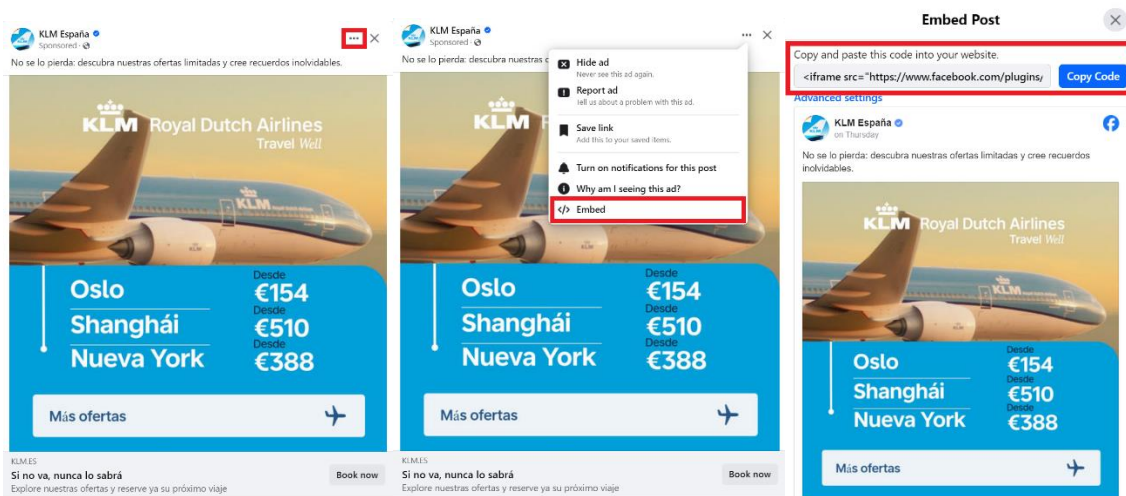


Illustration 13 – Inspection of HTML code on Facebook's ads

By exploring this option in both ads and posts, it was found that the “Embed” option appeared on a different position on the set of options, depending on whether it corresponded to an ad or to a post. It also varied even more depending on whether the content had images or videos, but the crucial finding here was that by determining the position of this embedding option, ads and posts could be successfully differentiated with a 100% accuracy. The exact position of this option will be detailed in the next subsection for all the different scenarios.

Illustration 14 shows the difference in position of the embedding option for an ad with image and for a post with image.

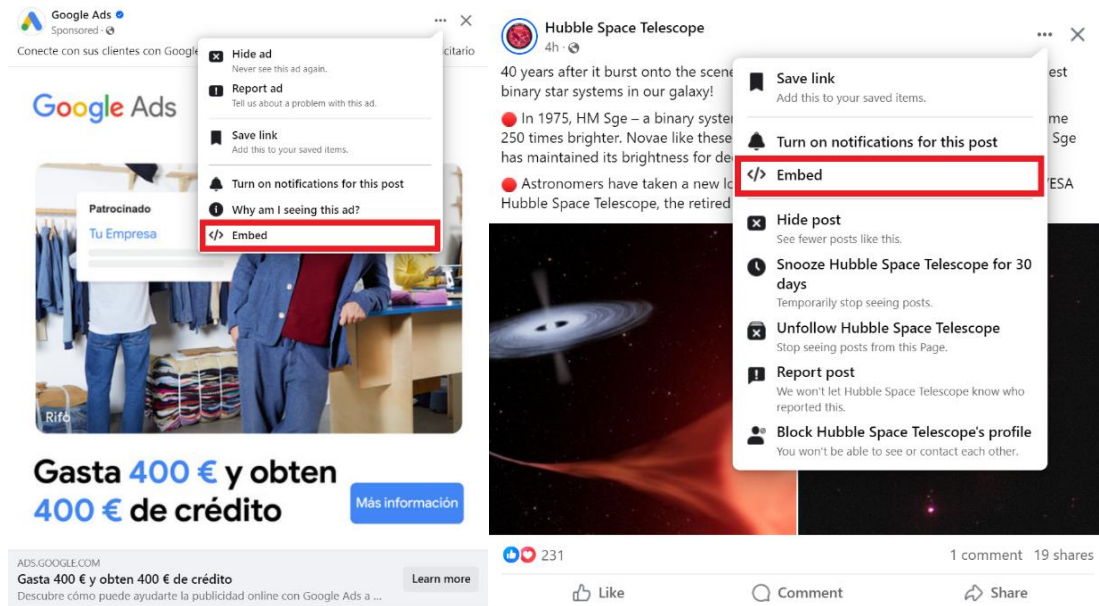


Illustration 14 – Location of the “Embed” option for ads (left) and posts (right) with images on Facebook

4.2.2 Scraping Software

The design of a scraper is extremely important in this work because the objective is to have an automatic collection of data that can be used to gather a large amount of ads and posts (which cannot be done manually) and observe the evolution of this subject in the future.

The scraping python program was created using a test automation software called Selenium, as it is the most complete tool that can perform all of the necessary tasks to accomplish this goal. This software is able to automatically replicate a series of actions that a human would perform on the web.

4.2.2.1 Ads Scraping

To start with, the necessary libraries are imported, including Selenium [16] and its functions and the web driver manager, which in this case corresponds to Google Chrome. It is decided to use this browser as it is easy to integrate it with Selenium, although the code written could be used for other browsers as well.

Also, there are other scraping alternatives to Selenium, such as Scrapy or Requests, but it was decided to proceed with Selenium because it was necessary to interact with

the elements in the page and Selenium is the most used scraping software for this type of problems, with constant updates and a variety of functionalities.

Then, the configuring options were defined for Chrome. This code was executed without using the common options such as using a headless browser for the purpose of visualizing every step of the scraping process to detect errors and improve its performance. But once it was optimized, it can be executed using any options for simplicity.

```
options = webdriver.ChromeOptions()
```

A new instance of the Chrome WebDriver is initialized in its latest version using selenium and proceeds to get Facebook's login page, maximizing its window for better visualization purposes.

```
driver = webdriver.Chrome(service=Service(ChromeDriverManager().install()),
options=options)

driver.get("https://www.facebook.com/login.php/")
```

Once the login page is loaded, the user's email and password need to be defined as strings and they need to be typed in their corresponding location. To ensure this, both the email and password input fields must be properly identified by defining their Xpath and then using the "send_keys" method, the user's login information is typed accordingly.

A stop timer was set after getting the login page (5 seconds) and in between typing inputs (2 seconds) to ensure that the action is performed on time and without any interruptions. These timers were determined by using trial and error methods without a headless browser. When the login information is already typed, the login button is identified and clicked on to enter the Facebook home page as an authenticated user.

```
password = 'type_password'

email = 'type_email'

time.sleep(5)

email_input = driver.find_element(By.XPATH, "///input[@name='email']")

email_input.send_keys(email)

time.sleep(2)

password_input = driver.find_element(By.XPATH, "///input[@name='pass']")
```

```
password_input.send_keys(password)

time.sleep(2)

log_in = driver.find_element(By.XPATH, "//button[@name='login']")

log_in.click()
```

Now, another stop timer (5 seconds) is set to wait for the home page to be loaded and then, the search for ads is initiated as well as the following ad parameters:

- UUID: using the “uuid” library [17], a random unique identifier is generated for each ad to be able to differentiate each one of them for future analysis.
- IsAd: boolean variable that determines if the scraped content corresponds to an ad or not. It should contain the value “Yes” for all of the entries of the generated data frame.
- Reactions: string variable that contains the number of reactions of each ad, such as likes or emoji reactions.
- Shares: string variable that contains the number of times the ad was shared.
- Comments: string variable that contains the number of comments.
- ContainsVideo: boolean variable that determines whether an ad contains a video or not.
- ContainsPicture: boolean variable that determines whether an ad contains an image or not.
- ContainsMedia: boolean variable that determines whether an ad contains either an image or a video or not. It should be mostly “Yes”, as ads without media are extremely rare.
- AdvertisingCompany: string variable that contains the name of the advertiser’s Facebook account.

A pandas [18] data frame is created to store all of the ad parameters above and then a loop is initiated to start scrolling down the feed. The number of scrolls is set to 1000, even though this number is unlikely to be reached due to Facebook’s limitations that will be discussed later.

```
for _ in range(1000):

    time.sleep(1)

    driver.execute_script("window.scrollTo(0, document.body.scrollHeight);")

    time.sleep(6)
```

In order to identify ads in the feed, the software first needs to identify all posts and ads shown after each scroll and then determine whether they are ads or not from the embedding option. The classes seen in this code may vary over time or due to the language set on Facebook, so they just need to be adjusted according to these factors.

```
posts = driver.find_elements(By.XPATH, '//div[@class="x1liihq"]')
```

A new loop is created to iterate over each of the posts found to collect all the data, starting with the UUID:

```
post_uuid = str(uuid.uuid4())
```

The code then proceeds to check the media variables, starting with “ContainsVideo”:

```
video_all = post.find_elements(By.XPATH, '//*[@playinline=""]')
```

```
contains_video = len(video_all) > 0
```

To check for the presence of images, there are 3 case scenarios to be considered, only if there were no video elements found in the previous condition. There can be just one image, multiple images or scrollable images which are identified in different ways, with try and except conditions to avoid errors:

```
if contains_video == False:

    try:

        image_element = post.find_elements(By.XPATH,
        '//*[@role="link"]//div//div//div//div//img') # one image

    except NoSuchElementException:

        image_element = 0

    try:

        multiple_images = post.find_elements(By.XPATH,
        '//*[@role="link"]//div//div//div//img') # multiple images

    except NoSuchElementException:

        multiple_images = 0

    try:

        scroll_images = post.find_elements(By.XPATH,
        '//*[@role="link"]//div//img[@alt="Image"]') # scrolling images
```

```

except NoSuchElementException:

    scroll_images = 0

    contains_picture = len(image_element) > 0 or len(multiple_images) > 0 or
len(scroll_images) > 0

    else:

        contains_picture = False

```

Once, the image and video variables are known, “ContainsMedia” can also be defined:

```
contains_media = contains_picture or contains_video
```

To collect the number of reactions, comments and shares, a try and except block has to be implemented for each of the parameters, as some of them may be missing if they are equal to 0.

```

try:

    reactions = post.find_element(By.XPATH,
'./div//span//div[@role="button"]//span[@aria-hidden="true"]//span//span').text

except NoSuchElementException:

    reactions = 0

try:

    comments = post.find_element(By.XPATH,
'./div//span//div[@id]//span[@dir="auto"]').text

except NoSuchElementException:

    comments = 0

try:

    shares = post.find_element(By.XPATH, './div[3]//span//div//span[@dir="auto"]').text

except NoSuchElementException:

    shares = 0

```

After getting all of the parameters, the embedding needs to be obtained using the preferred methodology explained in section 4.2.1, as well as determining which of the identified posts are ads. Therefore, the software locates the 3 dots on the top right corner, clicks on them, and adds a timer to wait for the different options to load.

The options displayed and their order vary depending on whether it is a post or ad, and whether it contains a video or not. So, the “Embed” options we need to identify to relate them to ads are the ones that appear on the 6th position for ads with no videos, and on the 7th position for ads with videos. Any other positions on the list for the option “Embed” would correspond to posts.

```

if contains_video == True:

    insertar_button = driver.find_element(By.XPATH, '//*[@div[7][@role="menuitem"]') # for
ads with videos, 'Insertar' is the 7th option

else:

    insertar_button = driver.find_element(By.XPATH, '//*[@div[6][@role="menuitem"]') # for
ads with no videos, 'Insertar' is the 6th option

```

Once the right option is identified, the software clicks on it and sets another timer to wait for the embedding menu to load.

```

if insertar_button.text == 'Insertar': #español

    is_ad = True

    insertar_button.click()

    time.sleep(7)

```

Depending on whether the ad has a video or not, the procedure to obtain the embedding content is slightly different. If it contains a video, before copying the HTML code a checkbox must be ticked to ensure that the whole content of the ad is loaded into the iframe, otherwise just the video would be included. On the other hand, ads with images already have the full embedding code ready to be copied. The method “get_attribute” is used to obtain the iframe code, and then the menu is closed.

```

if contains_video == True:

    checkbox = driver.find_element(By.XPATH, '//*[@div//input[@type="checkbox"]')

    checkbox.click()

```

```

time.sleep(4)

iframe_content = driver.find_element(By.XPATH, '//*[@aria-label="Entrada de
código de ejemplo"]').get_attribute('value') #español

time.sleep(7)

close_menu = driver.find_element(By.XPATH, '//*[@aria-label="Embed
Video"]/div[2]/div[@aria-label="Cerrar"]') #español

close_menu.click()

time.sleep(5)

else:

    iframe_content = driver.find_element(By.XPATH, '//*[@aria-label="Entrada de
código de ejemplo"]').get_attribute('value') #español

    time.sleep(7)

    close_menu = driver.find_element(By.XPATH, '//*[@aria-label="Insertar
publicación"]/div[2]/div[@aria-label="Cerrar"]') #español

    close_menu.click()

    time.sleep(5)

```

Lastly, the last parameter “AdvertisingCompany” is obtained, and the data frame is updated one ad at a time, while also adding a new column “Embedding” to store the iframe code. It is important to note that the whole iteration through the posts initially identified is enclosed in a try and except block to avoid errors and keep the code running.

This whole procedure was executed in a python environment using VSC, and a Windows operating system.

4.2.2.2 Posts Scraping

Due to the fact that identifying ads and posts is extremely similar to Facebook, it was very straightforward to adapt the code explained in section 4.2.2.1 to obtain a dataset of posts, without much additional effort. This code can obtain both ads and posts with the variable “IsAd” determining their category.

The login procedure and the collection of parameters is identical, but there are still a few differences in other steps. The first difference comes with the way of identifying which posts are ads, where the href element commented in 4.2.1 can be used.

For those posts with “IsAd” set to False, the same procedure is followed, with the small difference on the location of the “Embed” option. For posts with no videos, it is located in the 3rd position, while for ads with videos it is located in 4th position.

```

if contains_video == True:

    insertar_button = driver.find_element(By.XPATH, '//*[@div[4][@role="menuitem"]') # for
posts with videos, 'Insertar' is the 4th option

else:

    insertar_button = driver.find_element(By.XPATH, '//*[@div[3][@role="menuitem"]') # for
posts with no videos, 'Insertar' is the 3rd option

```

A fair question to be asked is why there are 2 different codes if ads and posts can be scraped by using just one piece of software. The answer to this question has to do with Facebook’s limitations on accounts and is developed in section 4.2.4.

4.2.3 Embeddings

After the scraping process, an iframe was obtained for each of the ads and posts, but in order to have a dataset of actual ads and posts, these iframes need to be saved into an HTML file. This phase was developed using a python environment in Google Colaboratory.

To start with, the data frames were uploaded and the entries with duplicate iframes were removed. Then, the iframes collected in column “Embedding” were saved into a new HTML file which is named after the UUID, in case an individual ad needs to be located afterwards. This procedure is done separately for ads and posts.

```

drive_directory = '/path_to_folder_containing_ads_or_posts'

if not os.path.exists(drive_directory):

    os.makedirs(drive_directory)

```

```

for index, row in df_facebook.iterrows():

    # Get unique identifier and iframe content

    uuid_value = df_facebook['UUID'][index]

    iframe_content = df_facebook['Embedding'][index]

    # Create html file with uuid value

    file_name = f"{uuid_value}.html"

    # Full path for each file

    file_path = os.path.join(drive_directory, file_name)

    # Write iframe to html file

    with open(file_path, 'w', encoding='utf-8') as file:

        file.write(iframe_content)

```

184 ads and 1414 posts were successfully obtained from Facebook for further analysis on their energy consumption.

4.2.4 Facebook Limitations

The reason why 2 different pieces of code were made for ads and posts is due to the restrictions imposed by Facebook. By performing the same action repeatedly over a period of time, i.e. clicking on the embedding option, Facebook temporarily blocks the Facebook account, usually for 24 hours, so only a small number of ads and posts can be obtained per execution of the code. This means that a real personal account should not be used for a long period of time as it is likely to be blocked. Illustration 15 shows the notification that a user gets when they get temporarily blocked by Facebook.

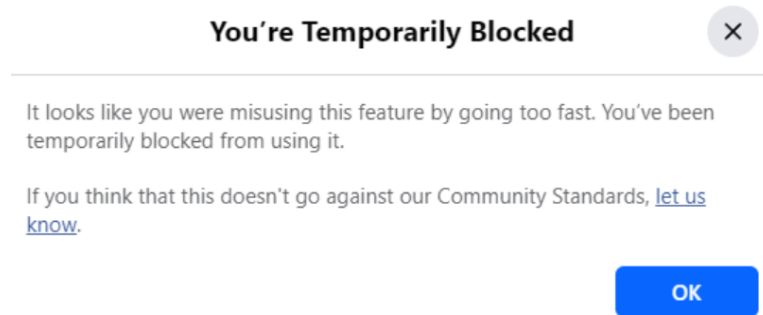


Illustration 15 – Notification of temporary account block on Facebook

As a result, a fake account was created and to register some activity, some famous accounts were followed. However, ads did not appear in this account's feed for a few months, so it was used only for getting posts.

The temporary block occurred after approximately 1 hour and 30 minutes, so to increase the amount of time that the code can be executed and retrieve more posts, a timer and execution cycles were implemented in the code. The best performance was obtained by executing 13 cycles of 15 minutes each, with a random pause between 60 and 90 minutes in between cycles. As a result, 314 posts were obtained before getting blocked.

Furthermore, another limitation imposed by Facebook is the embedding of some ads or posts. If their preview does not fully load in the embedding menu, it means that it is not eligible to be embedded. This can also be concluded if the ad's URI contains the following text: "https%3A%2F%2Fwww.facebook.com%2F100". The reason behind this is the privacy settings selected by advertisers or users that do not want their ad or post to be embedded anywhere. It is just a very small percentage of the total number of ads and posts (less than 5%), and nothing can be done to solve this. Illustration 16 represents the appearance of an invalid embedding on Facebook.

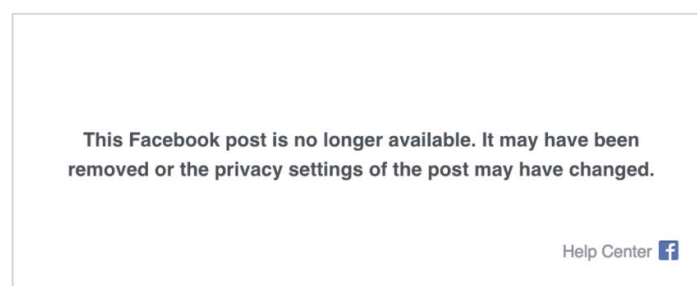


Illustration 16 – Appearance of invalid embedding

Lastly, ads that contain scrollable images do not contain the embedding option provided by Facebook, so they cannot be collected this way. This is not an important issue as the amount of ads with scrollable items that appear on the feed is very small.

4.3 DATA COLLECTION ON TWITTER

This section proceeds with the methodology to collect ads and posts on Twitter and obtain their embeddings.

4.3.1 Research Phase

The research phase on Twitter was very straightforward as the embedding option is also available for this social network, so the focus changed immediately to identifying the elements that differ between ads and posts.

Similarly to Facebook, ads contain an additional element that visually tells that there is an ad. In this case, it shows the word “Ad” on the top right corner of the ad.



Illustration 17 – Visual difference between ads and posts on Twitter

4.3.2 Scraping Software

Again, the scraping software created for Twitter follows the same idea as the one for Facebook, with a few adaptations to make it suitable for this social network. However, for the embedding procedure, it is quite different from Facebook as it has to be done on a completely new page, so it is done separately once the scraping process is finished.

To start with, the web driver is defined identically to the one in section 4.2.2. Then, the authentication is done, also in the same way as in Facebook, but changing the HTML elements identified on Twitter.

```
username = 'type_username'

password = 'type_password'

time.sleep(5)

username_input = driver.find_element(By.XPATH, "//input[@name='text']")

username_input.send_keys(username)

time.sleep(5)

siguiente = driver.find_element(By.XPATH, "//span[contains(text(),'Next')]")

siguiente.click()

time.sleep(5)

password_input = driver.find_element(By.XPATH, "//input[@name='password']")

password_input.send_keys(password)

time.sleep(5)

log_in = driver.find_element(By.XPATH, "//span[contains(text(),'Log in')]")

log_in.click()

time.sleep(5)
```

It is important to note that a one-time authentication was required by using the email, as a way of confirming that the device was reliable. A code received via email had to be typed manually to be able to log in.

The data frame to be generated has very similar parameters to the Facebook one, with a few differences to be noted. Twitter shows the amount of “Views” and “Bookmarks” for every post and ad, while the number of reactions is not shown. Also, the column “Shares” is renamed to “Reposts”, and a new one called “URL” is added, which stores the URI of the post or ad.

The complete list of parameters collected for Twitter ads and posts are the following:

- UUID: using the “uuid” library [17], a random unique identifier is generated for each ad/post to be able to differentiate each one of them for future analysis.
- IsAd: boolean variable that determines if the scraped content corresponds to an ad or not.
- Likes: variable that contains the number of likes of each ad/post.
- Views: variable that contains the number of times users see each ad/post.
- Bookmarks: variable that contains the number of times an ad/post was saved.
- Reposts: variable that contains the number of times the ad/post was reposted.
- Comments: string variable that contains the number of comments.
- ContainsVideo: boolean variable that determines whether an ad/post contains a video or not.
- ContainsPicture: boolean variable that determines whether an ad/post contains an image or not.
- ContainsMedia: boolean variable that determines whether an ad/post contains either an image or a video or none.
- AdvertisingCompany: string variable that contains the name of the advertiser’s Facebook account. Returns “None” if it is a post.
- URL: variable that contains the URL of the ad/post, extracted from the HTML code of the page, that will be used to obtain embeddings in the next step.

Once Twitter’s home page is loaded, the number of posts and ads per scroll is identified, and then the ones that are ads are differentiated by locating the text “Ad”. In addition, the URI of each of them is extracted from the HTML code to get the embedding during the last phase, identified by finding the text “/status/” in an href element.

```

posts = driver.find_elements(By.XPATH, '//div[@data-testid="cellInnerDiv"]')

for tweet in posts:

    post_uuid = str(uuid.uuid4())

    post_url = tweet.find_element(By.XPATH, '//*[@contains(@href,
"/status/")]').get_attribute("href")

    try:

        is_ad_element = tweet.find_element(By.XPATH,
'//*[@dir="ltr"]//span[contains(text(), "Ad")]')

        is_ad = is_ad_element.text == "Ad"

    except NoSuchElementException:

        is_ad = False

```

Furthermore, the interaction parameters are identified, this time by using regular expressions, as they all appear together in a string and need to be separated. It is important to realize that some of these parameters might not be present for all posts and ads, meaning that they are equal to 0, so this condition must be included to avoid any errors.

```

aria_label = tweet.find_element(By.XPATH, '//*[@role="group"]').get_attribute('aria-label')

likes_match = re.search(r'(\d+)\s+likes?', aria_label)

likes = int(likes_match.group(1)) if likes_match else 0

reposts_match = re.search(r'(\d+)\s+reposts?', aria_label)

reposts = int(reposts_match.group(1)) if reposts_match else 0

bookmarks_match = re.search(r'(\d+)\s+bookmarks?', aria_label)

bookmarks = int(bookmarks_match.group(1)) if bookmarks_match else 0

views_match = re.search(r'(\d+)\s+views?', aria_label)

views = int(views_match.group(1)) if views_match else 0

comments_match = re.search(r'(\d+)\s+replies?', aria_label)

comments = int(comments_match.group(1)) if comments_match else 0

```

Illustration 18 shows the “div” element of a Twitter ad that contains the information to be extracted about the interaction parameters.

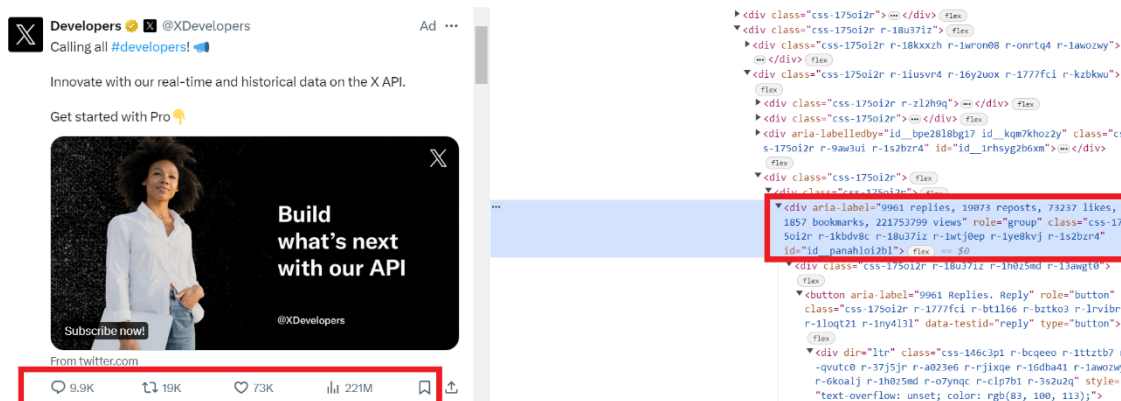


Illustration 18 – Location of Twitter’s interaction parameters in HTML code

Moreover, the parameters related to the visual elements and the name of the advertisers, if applicable, are collected. The video elements on Twitter also include GIFs and the image elements are either images with or without a link attached to it.

```
video_all = tweet.find_elements(By.XPATH, '//*[@data-testid="videoPlayer"]')
```

```
contains_video = len(video_all) > 0
```

```
if contains_video == False:
```

```
image_element = tweet.find_elements(By.XPATH, '//*[@data-testid="tweetPhoto"]')
```

```
image_general = tweet.find_elements(By.XPATH, '//*[@data-testid="card.wrapper"]') #link
```

```
contains_picture = len(image_element) > 0 or len(image_general) > 0
```

else:

```
contains_picture = False
```

```
contains_media = contains_picture or contains_video
```

```
advertising_company = ""
```

if is ad:

```
advertising_company = tweet.find_element(By.XPATH, '//*[@data-testid="User-Name"]//div[@dir="ltr"]').text
```

To end with, all values are updated into the data frame, and the try and except block which encloses the whole loop is ended.

4.3.3 Embeddings

Once the scraping software has ended, a dataset of parameters is obtained, including the “URL”, which is used in this next phase to obtain the embedding of ads and posts. To do this, a new automated software was designed that works with the dataset of URLs as input and generates an HTML file containing the embedding of every ad and post.

Before executing this, the URLs need to be cleaned by removing a fragment of its text that is not used to generate the embedding, as well as removing the duplicate entries of the dataset.

```
df_twitter['URL_cleaned'] = df_twitter['URL'].str.replace(r'/analytics$/photo/1$', '')
```

Basically, the procedure consists of opening a new browser for every entry of the dataset and visiting the webpage “<https://publish.twitter.com/>”, where the cleaned URL is typed. Illustration 19 represents the first step of the embedding process, where the URL of the ad or post has to be introduced.

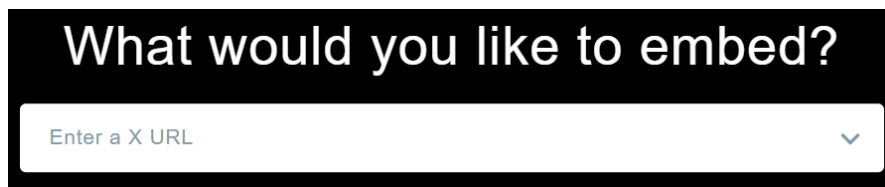


Illustration 19 – Input of the URL code extracted from Twitter

Then, two options appear, either embedding a post or embedding a video, and depending on the elements of each ad or post, the corresponding option is clicked. Illustration 20 presents the second step in the embedding process, where the embedding content has to be chosen adequately.

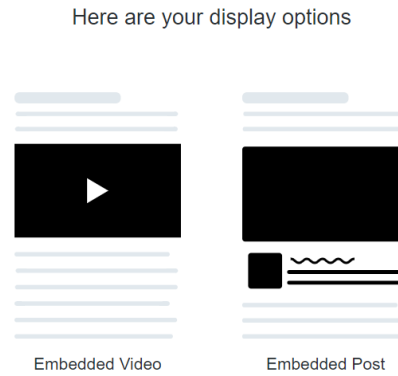


Illustration 20 – Embedding options

Lastly, a preview of the ad or post is shown and the HTML code for the embedding is copied and saved into an HTML file with its name corresponding to its UUID. Illustration 21 shows the third step to embed an ad or post, where their HTML content can be copied.



Illustration 21 – Preview of generated embedding

The code to achieve this is displayed below, executed in Google Colaboratory using a headless browser:

```

drive_directory = 'path_to_folder_containing_ads_and_posts'

if not os.path.exists(drive_directory):

    os.makedirs(drive_directory)

for index, row in df_twitter_ads.iterrows():

    time.sleep(5)

    options = webdriver.ChromeOptions()

    driver = webdriver.Chrome(options=options)

    driver.get("https://publish.twitter.com/")

    time.sleep(5)

    try:

        url = row['URL_cleaned']

        uuid_value = row['UUID']

        url_input = driver.find_element(By.XPATH, "///input[@class='WidgetQuery-input']")

        url_input.send_keys(url)

        time.sleep(5)

        try:

            button_url = driver.find_element(By.XPATH, "///button[@class='WidgetQuery-button
WidgetQuery-submit Icon Icon--arrowRight']")

            button_url.click()

        except ElementClickInterceptedException:

            button_url = driver.find_element(By.XPATH, "///button[@class='WidgetQuery-button
WidgetQuery-submit Icon Icon--arrowRight']")

            driver.execute_script("arguments[0].click();", button_url)

        time.sleep(3)

        try:

            embed_post = driver.find_element(By.XPATH, "///span[@class='WidgetsSelector-image
Icon Icon--tweetSelector']")

```

```

        embed_post.click()

    except ElementClickInterceptedException:

        embed_post = driver.find_element(By.XPATH, "//span[@class='WidgetsSelector-image
Icon Icon--tweetSelector']")

        driver.execute_script("arguments[0].click();", embed_post)

        time.sleep(5)

        iframe_content = driver.find_element(By.XPATH, "//code[@class='EmbedCode-
code']").text

        time.sleep(5)

        file_name = f"{uuid_value}.html"

        file_path = os.path.join(drive_directory, file_name)

        with open(file_path, 'w', encoding='utf-8') as file:

            file.write(iframe_content)

        print(f"HTML File '{file_name}' generated for URL: {url} with index {index} at
'{file_path}'")

    except NoSuchElementException:

        print(f"It is not possible to get the iframe content for URL: {url} with index {index}")

```

324 ads and 5355 posts were successfully obtained by following this procedure.

4.3.4 Twitter Limitations

Unlike Facebook, Twitter imposes less limitations to maximize the amount of ads and posts to be scraped. One of the things that need to be considered is that once a new account is created, it takes around 1 week for ads to start showing in the account's feed. As found in the account settings, the amount of audience lists that the account is part of and the number advertisers targeting the account is shown in Illustration 22.

← Your advertiser list

Tailored audiences are often built from email lists or browsing behaviors. They help advertisers reach prospective customers or people who have already expressed interest in their business. [Learn more](#)

You are currently a part of **0 audiences** from **0 advertisers**

You can opt out of interest-based advertising in your personalization and data settings. This will change the ads you see on X, however it won't remove you from advertisers' audiences.

Illustration 22 – Settings view of advertiser lists on Twitter

Another limitation is that the number of ads and posts shown on Twitter is limited, as there are not infinite ads to be displayed to a user, and the number of scrolls that Twitter allows over a period of time is also limited. Also, the variety of posts is not infinite either, so even though the software is executed many times, the same ads are likely to appear, until they are updated after several days.

Nevertheless, the developed software can still obtain a large quantity of ads and posts, approximately 6,000 per execution and per day. This is extremely useful because the X (Twitter) API only allows users to retrieve 1,500 tweets for free and 3,000 tweets with the basic subscription that costs 100\$ per month [19]. Therefore, by executing this software once, twice the amount of tweets can be obtained than the paid API.

4.4 MEASURING ENERGY CONSUMPTION

After obtaining 4 different datasets, one per combination of social network (Facebook or Twitter) and type of content (ad or post), the next step consists of measuring the energy consumed by each one of the HTML files across these 4 datasets, as well as other parameters that can be useful for analyzing the results.

4.4.1 Measurement Methodology

In order to measure the energy consumption of a piece of code, literature shows a scientific method validated using a library called CodeCarbon [20]. Researchers of the mentioned work implemented a methodology to measure the energy consumption of online ads in the programmatic ecosystem. For this purpose, the software developed loads each one of the HTML files containing the ad or post content on a clean browser alone. It computes the baseline energy of the computer and browser itself by doing a set

of different baseline measurements. This way the code developed by the researchers ensures that the amount of energy used by the piece of HTML loaded corresponds uniquely to the HTML itself and not the baseline energy consumed by other processes in the computer, such as the browser [13].

This work had access to the package (not publicly available) that applied the methodology explained above by the researchers. With that, the following parameters were collected:

- Loading time: time it takes to load each ad or post.
- Energy consumption: estimation of the amount of energy consumed (in kWh), which is computed as the sum of the energy used per CPU, GPU, and RAM in the device.
- Length of entries: number of internal calls needed by the ad or post

It is very important to note that the energy measurements were always conducted using the same network connection in the United Arab Emirates, to avoid any energetic impact derived from the connection.

4.4.2 Limitations of the Measurements

During the measurement period, some limitations were found. The execution of this code was quite time-consuming as measuring the parameters of each HTML file took between 15 and 20 seconds. Therefore, only one electronic device was used in this work, a Microsoft Surface laptop with the following characteristics:

- Installed RAM: 8.00 GB
- Processor: Intel(R) Core(TM) i5-8250U CPU @ 1.60GHz 1.80 GHz
- System type: 64-bit operating system, x64-based processor
- OS: Windows

Nevertheless, using just one device does not influence the comparison of results, as the comparisons made in this work are between Facebook and Twitter and ads and posts, and not between different devices.

5. DATASET

After the data collection phase was completed, a total of 7,277 HTML files were successfully retrieved along with 11 parameters for those corresponding to Facebook, and 12 parameters for those corresponding to Twitter. These files are distributed in the following way, depending on the social network and type of content:

Table 1. Distribution of ads and posts on Facebook and Twitter

	FACEBOOK	TWITTER	TOTAL
ADS	184	324	508
POSTS	1,414	5,355	6,769
TOTAL	1,598	5,679	7,277

The data was extracted from Twitter between the 16th of December 2023 and the 25th of January 2024, and from Facebook between the 19th of February 2024 and the 5th of March 2024. Besides, the energy measurements were conducted between the 7th of March 2024 and the 5th of April 2024.

6. RESULTS

The next section is intended to give a well-founded answer regarding the factors that influence the energetic cost of ads and posts, as well as the differences in energy consumption between ads and posts within Facebook and Twitter and among these two social networks.

6.1 IMPORTANT FACTORS

One of the crucial questions about the energy consumption of ads and posts is to understand the elements that are related to a greater consumption. For this reason, in this first subsection the 184 ads and 1,414 posts from the Facebook dataset and 324 ads and 5,355 posts from the Twitter dataset have been taken together with each one of the attributes associated with them.

These datasets were explored and preprocessed to be ready for the analysis phase, by removing the unnecessary columns and outliers, and converting the variables to the proper data types.

The following figures presented in this subsection depict the correlation matrices of the different variables that were obtained during the data collection phase, to determine which variables have more influence on energy consumption.

Figure 3 presents two correlation matrices that show the energy consumption against the rest of the variables previously collected, such as “Reactions”, “Shares”, or “ContainsVideo”, for ads and for posts separately. The darker colors suggest a stronger correlation between variables, while the lighter colors indicate that there is very small correlation. Also, the colors with a red tone mean that the correlation is positive while the colors with a blue tone suggest a negative correlation.

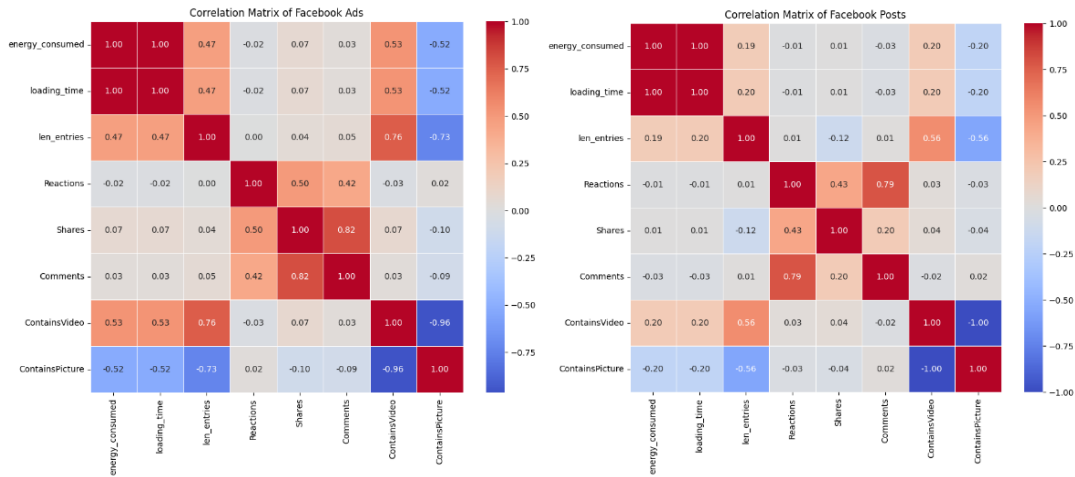


Figure 3 – Correlation matrices of the parameters on Facebook ads (left) and posts (right).

As it can be seen from Figure 3, where the correlation matrices for both ads and posts on Facebook are plotted, the energy consumption and the loading time are perfectly correlated. This makes complete sense as the amount of time taken to load an ad or post will linearly increase relative to the amount of energy it takes for the device to load the ad. In fact, this perfect correlation can be found for all datasets.

In addition, a correlation can be seen between the variable “len_entries” and the energy consumption in both Facebook ads (correlation of 0.47) and posts (correlation of 0.19). This suggests that there is a relationship between the number of internal calls needed to load and ad or post with the amount of energy being consumed during that process. This relation is stronger for ads, suggesting that more events happen in the background probably due to tracking the user.

Also, it can be observed that for Facebook ads that contain a video, the energy consumption increases (correlation of 0.53) while for ads that contain images, the opposite occurs (correlation of -0.52). For posts, this relationship is also present, but it is less relevant as the correlation value is significantly lower.

One possible explanation for this is the fact that online ads require to load more resources on the background to check the view of the image or the video and send these parameters to Facebook so it can later charge users accordingly. However, that is not needed for the case of posts.

Moreover, videos may load much more content in the rendering process in comparison to images, and that is a feasible explanation of the differences in this correlation.

Furthermore, the rest of variables, which are the interaction parameters including reactions, shares, and comments, are proven to be insignificant to the energy consumption of Facebook ads and posts. This makes complete sense as they do not influence in any way the loading of the ad or post, and therefore, serves as validation of the correctness of the measurements. The energy consumption is only influenced by parameters related to the loading of the ad or post and not by their popularity.

Figure 4 also represents the correlation matrices of energy consumption with the rest of the variables, for ads and posts, but this time for Twitter.

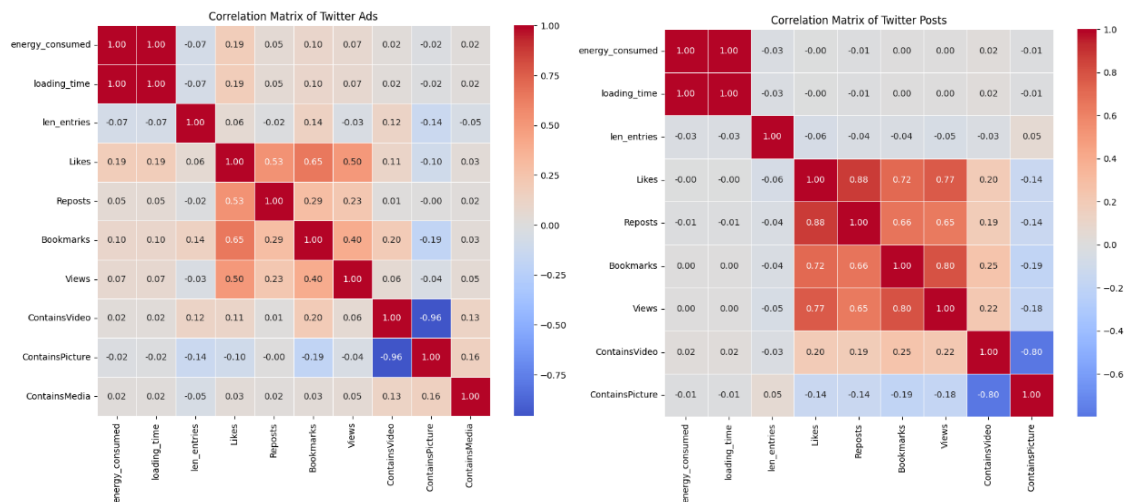


Figure 4 – Correlation matrices of the parameters on Twitter ads (left) and posts (right).

As seen above in Figure 4, the presence of images and videos does not influence at all the consumption of energy in neither ads nor posts. However, for ads, the interaction parameters, especially the number of likes (correlation of 0.19), are slightly positively correlated with the energy consumed, while for posts this relationship does not exist. It was expected that, as in the case of Facebook, interaction variables would not affect the energy consumption. However, this light correlation suggests that popular ads have a higher energetic cost, maybe due to the possibility that during its loading process, some extra information is reported to position the ad as relevant, but this is just a hypothesis and it cannot be proven.

In this case, there is no significant relationship between the number of internal requests (“len_entries”) and the amount of energy consumed by Twitter ads and posts. This suggests that Twitter does not rely on additional calls related to users or ad tracking to load their content on the feed.

The main insights reported here are that energy consumption of ads and posts is related to the loading time, the number of internal calls, and the display of video or image in the case of Facebook and only to the loading time in the case of Twitter, regardless of the type of content.

6.2 ADS VS POSTS ON FACEBOOK

Another interesting question is to understand if it exists a relevant difference between the energy consumption of ads and posts. If this difference existed, it may reveal a crucial insight on energy consumption of social networks, implying that energy consumption associated to users’ behavior is directly related to the amount of posts or ads shown to them.

For that reason, the difference in energy consumption and loading time between ads and posts on Facebook is analyzed in depth, which is crucial to finding any underlying relationships hidden in the data. By understanding these relationships, it is possible to confidently differentiate the more energy consuming contents in a user’s feed.

Figure 5 depicts two box plots. On the left plot, the y-axis represents the distribution of energy consumed by ads and posts on Facebook, differentiated in the x-axis. On the right plot, the y-axis represents the distribution of the loading time of ads and posts Facebook. They show two different results per plot, one for Facebook ads and another one for Facebook posts. The two plots (left and right) have identical shapes due to the perfect correlation between the variables “energy_consumed” and “loading_time”.

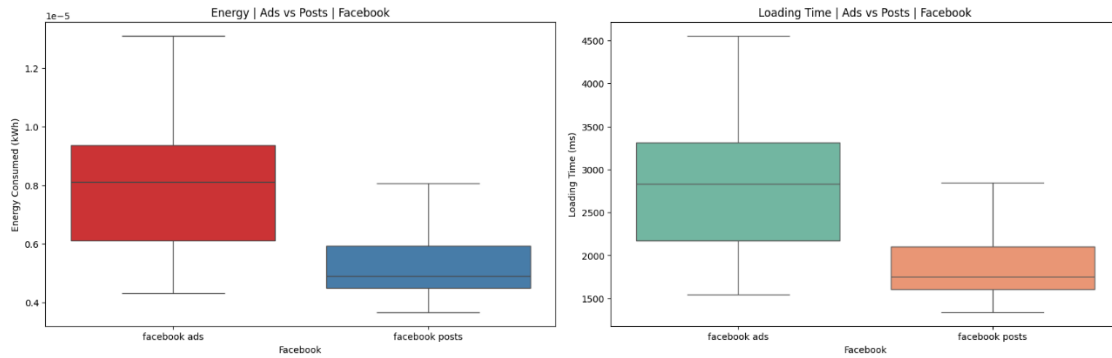


Figure 5 – Box plots of Facebook ads and posts vs energy consumed (left) and loading time (right)

Figure 5 shows a really striking result. Not only it depicts the difference between ads and posts but it shows an incredible difference in the energy consumed by ads in comparison to posts. It can be seen that the energy consumption of Facebook ads is much higher than Facebook posts, with a median of approximately 0.8×10^{-5} kWh for ads and 0.5×10^{-5} kWh for posts. This means that the consumption of ads is about 1.6 times higher than posts.

In addition, it is shown that the loading time of Facebook ads is much higher than on Facebook posts, with a median of around 2.83 seconds for ads and 1.75 seconds for posts. This suggests that the rendered content is different (e.g., higher size or number of calls) and requests needed to load an ad on Facebook are much slower than in posts, needing about 1.6 times the amount of time.

These are both outstanding findings as it is proven that ads on Facebook consume significantly more energy than posts, and require more loading time, which is extremely relevant in order to increase awareness of the amount of energy that is being consumed with every action on Facebook.

6.3 ADS VS POSTS ON TWITTER

For Twitter, the same visualization was done, as shown below in Figure 6. On the left plot, the y-axis represents the distribution of energy consumed by ads and posts, differentiated in the x-axis. On the right plot, the y-axis represents the distribution of the loading time of ads and posts. They show two different results per plot, one for Twitter ads and another one for Twitter posts.

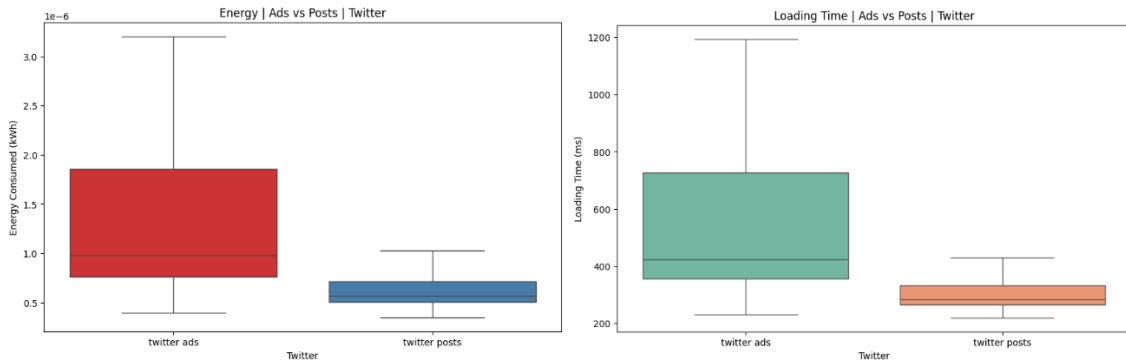


Figure 6 – Box plots of Twitter ads and posts vs energy consumed (left) and loading time (right)

The results found in Figure 6 are astonishing. The median energy consumption is around 1.0×10^{-6} kWh for Twitter ads and 0.6×10^{-6} kWh for Twitter posts. This means that the consumption of ads on Twitter is almost 1.75 times higher than posts.

Moreover, the median loading time was found to be around 0.42 seconds for ads and 0.28 seconds for posts, meaning that ads need about 1.5 more times the amount of time that posts need to be loaded.

These results, like the ones for Facebook, represent very interesting discoveries that suggest that ads require more, or heavier, elements or calls to be loaded onto the page than posts, therefore making them significantly more energy consuming.

6.4 FACEBOOK VS TWITTER

Consequently, it is very important to analyze the results obtained in the previous sections but from a different perspective, this time across social networks. This is very important to be able to tell whether content on Facebook or Twitter is more energetically expensive.

It has been found that ads consume more energy than posts in both social networks, but which social network consumes more? And which one is more optimized based on their consumption ratio?

Figure 7 represents the answer to the first question posed above. On the left column, the distribution energy consumption of posts (top) and ads (bottom) is plotted on the y-axis and differentiated into Facebook and Twitter in the x-axis. On the right column, the distribution of the loading time is plotted on the y-axis and differentiated into social networks on the x-axis.

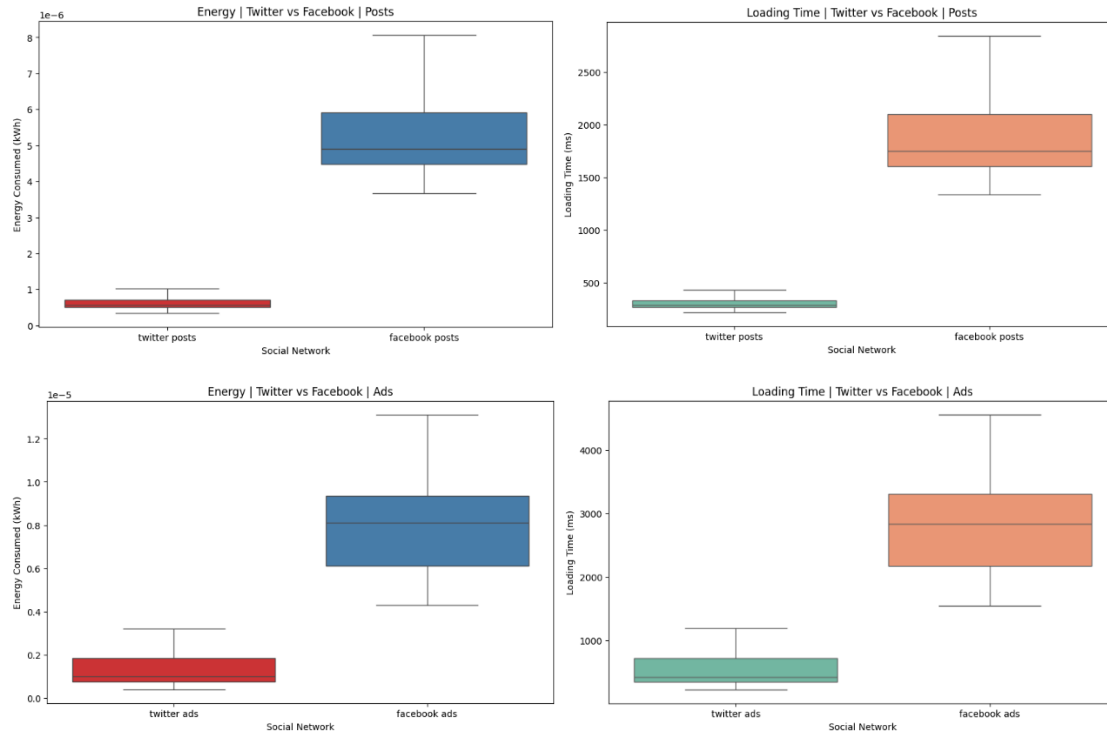


Figure 7 – Boxplots of the energy consumed (left) and loading time (right) vs social network

Figure 7 shows very interesting results. In terms of energy consumption, it is demonstrated that Facebook consumes around 8.3 times more energy than Twitter in posts and 8.7 times more in ads. Regarding the loading time, Facebook takes 6.2 times more the duration it takes for Twitter to load posts and 6.7 times more for ads.

These findings indicate that Facebook's platform is more energy consuming than Twitter's, which suggests that there are more events happening on the background to load ads and posts into Facebook's page, maybe due to the tracking of the ad or user, which is something that does not occur on Twitter on the same scale.

However, even though Facebook is more energy consuming than Twitter due to the longer time it takes to load ads and posts, it is very interesting to analyze the optimization ratio that these 2 platforms have between the energy consumption and the loading time.

Figure 8 illustrates the ratio of energy consumption to loading time of ads and posts in both Facebook and Twitter. This ratio is relevant because it depicts the amount of energy consumed per millisecond of loading time and represents the optimization of this process.

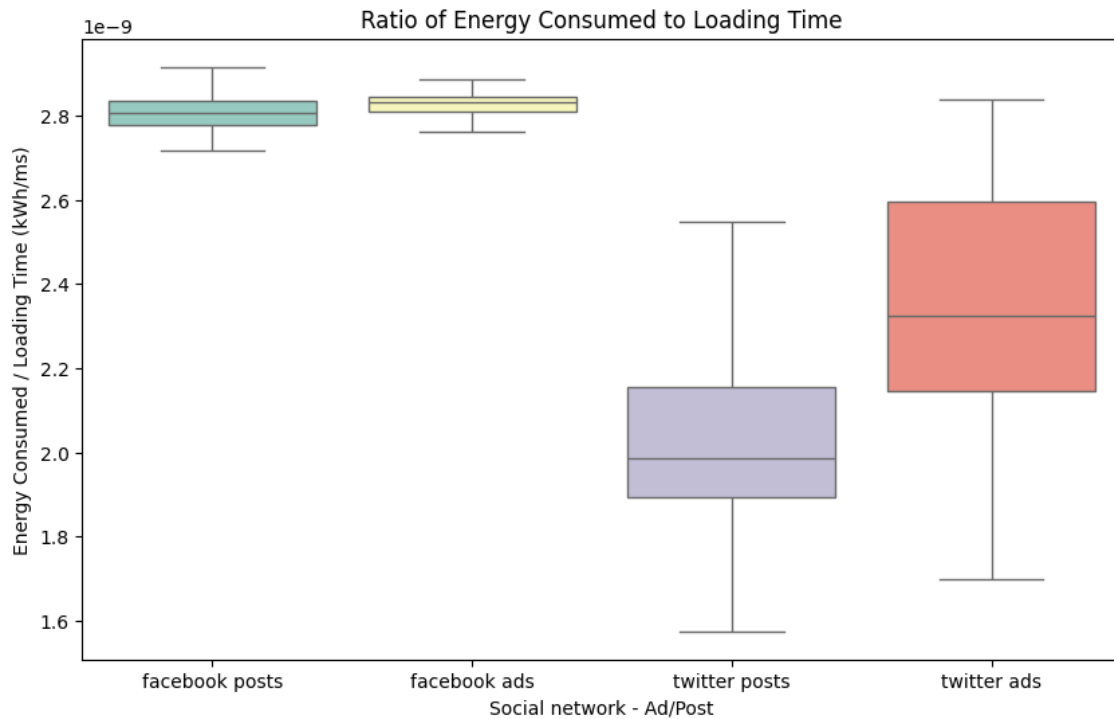


Figure 8 – Ratio of energy consumption vs loading time of ads and posts on Facebook and Twitter

As shown in Figure 8, Twitter is better optimized in the ratio of energy consumed versus loading time, in both ads and posts. This means that for every millisecond of loading time, Facebook consumes approximately 1.2 times more energy than Twitter in ads and 1.4 times more in posts.

In addition, ads still have a higher energetic cost ratio, which is more perceptible on Twitter, but is also present on Facebook even though it is smaller.

The results in this section indicate that Twitter is a greener social network than Facebook as it is less energy consuming, but also is more optimized in terms of energy consumption per millisecond of loading time, both for ads and posts. This strongly suggests that Facebook requires additional internal calls every time an ad or post is uploaded onto their social platform and due to these calls, more energy is spent by the users that are visualizing their content.

This information is therefore critical for users to wisely choose which platforms to use and how much time to spend using them considering the amount of energy that the user's device would consume.

6.5 DISCUSSION

It is very important to give some context to the results obtained in the sections above to better understand their influence and impact. Therefore, by taking into consideration the correlation matrices in 6.1, the median values obtained from the box plots in sections 6.2-6.4, and the data explained in section 1, the extent of the results can be better interpreted.

The correlation matrices corresponding to Figures 3 and 4 clearly indicate that there are some factors that influence the energy consumption of both ads and posts, depending on the social network they belong to. The loading time of the ad was proven to be the most significant factor across both social networks, by having a perfect positive correlation with the energetic cost.

For Facebook, the most influencing factors (besides loading time) are the content type of the ad or post, as the ones containing videos contribute to a higher energy consumption, while content with images have the opposite effect. This relationship has a higher weight in ads than in posts, as they are likely to require more internal calls to load videos onto the page than regular posts, which relates to the second most significant factor on Facebook. The amount of internal calls required for an ad or post to load are also a determining factor to energy consumption, being more important for ads, as these calls might be more present due to ad or user tracking parameters.

On the other hand, the energy consumption of Twitter ads, and not posts, (besides loading time) is only slightly affected by some interaction parameters such as the number of likes and bookmarks. These variables suggest that there is a small correlation between the popularity of the ad and the energy being consumed by it, maybe due to some additional information being added during its loading process to position the ad as relevant, but this is just a hypothetical explanation.

Moreover, the distribution of ads and posts were compared within each social network to understand the difference in energy consumption and loading time depending on the type of content that is shown in the feed. For Facebook, it was found that both the energy consumed and loading time of ads is about 1.6 times higher than posts. For

Twitter, the difference in energy consumption was even higher, around 1.75 times more for ads than for posts, but lower in loading time, 1.5 times more for ads.

These differences suggest that both platforms are using more resources to load their ads onto their feed than the ones they are using for posts. These additional resources might be related to ad or user tracking, such as popularity parameters.

Coming back to the data introduced at the beginning of this work, as seen in Figure 2, Facebook is the second most energy consuming social network among the 10 selected ones, while Twitter is in 6th position. This fact seems logical taking into account the number of users on each platform, as Facebook has many more users than Twitter.

However, as seen from Figure 7, Facebook consumes around 8.3 times more energy than Twitter in posts and 8.7 times more in ads, and takes 6.2 times more the duration it takes for Twitter to load posts and 6.7 times more for ads. Also, Figure 8 indicates that the ratio of energy consumption and loading time is also higher for Facebook in both ads and posts, consuming 1.2 times more energy per millisecond of loading in ads, and 1.4 times more in posts.

Together, these striking findings imply that Twitter is a greener and more optimized social platform in terms of energy consumption and that any user would be consuming less energy by using their platform than by using Facebook, and due to this a user might want to shift their use to become more energy efficient.

Overall, the results obtained present an ethical dilemma to advertisers and the platforms themselves, suggesting they should work together towards reducing the amount of energy being spent while they are making profit. If they can reduce the energy consumed by ads to a similar amount as posts, a huge amount energy would be saved in the long run.

Furthermore, if the advertising reach of each platform and the median amount of energy consumed per ad are considered, it is possible to make some rough estimations. In the fourth quarter of 2023, Facebook reported an average of 2.11 billion daily active users [21], and if every user were to view a median of 70 ads per week, as stated by researchers [22], (5.6×10^{-5} kWh per user every week), the estimated amount of energy in median used per week would be around 1.1 million kWh, or almost 62 million kWh per year. This amount is almost the equivalent of the electric energy consumption of a small country such as Vanuatu, which consumed approximately 65 million kWh in 2018, based

on the EIA [23]. This is an appalling comparison, considering this is just the estimated amount for one social network, and the number of users is constantly increasing.

Twitter has a smaller estimated audience is much less than Facebook's, but it still has 245 million daily active users worldwide [24]. If every user were to see 70 ads per week as on Facebook (0.7×10^{-5} kWh per user per week), ads on Twitter would consume around 140,000 kWh per week and 7.1 million kWh per year if every user were to visualize one ad per day. This is significantly less than Facebook, but it is still a huge amount, as it is almost equal to the energy consumed by around 1,155 residential electric-utility customers per year in Hawaii in 2022 (6,178 kWh each per year), according to the EIA [25]. This amount is still significantly high and should be able to inform users of the effect of their usage in this social network.

7. REGULATORY FRAMEWORK

Recently, an energy crisis has arisen in Europe because of the Russo-Ukrainian war [26], which led to the cease in imports of fossil fuels from Russia [27]. This has significantly raised the price of energy-related services, such as electricity in households [28].

7.1 EU REGULATIONS

As a result, the European Union (EU) has adopted new laws and regulations such as the ones detailed in the “Directive 2023/1791 of the European Parliament and of the Council”. This recent directive describes the key elements in Europe’s energy transition and focuses on the principle of “energy efficiency first” [36], meaning that all EU countries must consider from now on the energy efficiency aspects in all policy and investment decisions across all sectors.

Mainly, this directive imposes on EU member states an action plan to ensure a reduction in energy consumption by at least 11.7% of the current level in 2030. As a result, the overall energy consumption in the EU should not exceed 763 million tons of oil equivalent (Mtoe) by 2030.

To this end, there is also an obligation to EU member states to achieve certain levels of annual energy savings of at least 1.3% in 2024-2025, 1.5% in 2026-2027, and 1.9% in 2028-2030.

Furthermore, this directive also recalls the importance of energy saving in different sectors, highlighting that up to 9% of the energy consumption worldwide comes from the information and communications technology sector [37].

In addition, it tackles the issue of energy poverty, as 75% of the EU’s buildings have poor energy performance. It focuses on energy efficiency improvements for vulnerable customers, individuals affected by energy poverty, and those living in social housing.

The Delegated Regulation (EU/2024/1364) [38] also enforces the monitoring of energy performance of data centers, which accounted for almost 3% of the total energy consumption in the EU in 2018. This regulation aims to rate the sustainability of data centers in the EU based on the information and key performance indicators communicated to the European database.

Moreover, the REPowerEU Plan [39], adopted in May 2022 as a result of Russia's invasion of Ukraine, aims to increase the transition rate towards a cleaner and more resilient energy system in the EU.

Some examples of energy saving ongoing projects in member states after the creation of this plan include the following:

- France is supporting the energy renovation of almost 1.8 million households, 40,000 social housing units, and 5,000 small enterprises, as well as the thermal renovation in thousands of public sites and buildings [39].
- Austria is supporting thermal renovation with the launch of hundreds of projects to increase the efficiency of heating, contributing to improve the energy poverty crisis [39].
- Greece is working on increasing the energy efficiency of thousands of households and installing more than 170,000 solar water heating systems [39].

7.2 GLOBAL REGULATIONS

The energy directives imposed by the EU are in line with the ultimate global objective towards energy efficiency. On December 2023, the Conference of the Parties (COP28) was celebrated and representatives of more than 100 countries along with members of the UN agreed on the launch of the “Global Pledge on Renewables and Energy Efficiency” [40].

This global initiative aims to double the yearly rate of global energy efficiency improvements from 2% to 4% by 2030 and to triple the installed capacity of renewable energy. The EU is planning to invest 2.3 billion euros from the EU budget to support this energy transition program, create green jobs and invest in technologies of the future.

7.3 NEXT STEPS

There has been some progress in terms of legislation towards a new era with renewable energy, energy efficiency, and energy saving. However, the current regulations and objectives fail to address the outrageous amount of energy being consumed by online advertising and the use of social networks.

As today's society transitions to a greener energy system, it is also transitioning to a digitalized environment, where the energy consumed by digital devices is increasing as the number of users increases. Due to this, the energy consumption from ads on social networks is going to increase accordingly, and this presents a huge problem, as the results from this work have shown.

To address this issue, new and specific regulations need to be studied and imposed by the corresponding parties, which would even add value to the existing legislation explained above and contribute to a faster and smoother transition to energy efficiency.

7.4 CONTRIBUTION OF THIS WORK

The topic and results of this work are of extreme importance to the development of new laws and regulations regarding the energy consumption of technological processes. It has unveiled a new dimension to the problem of energetic cost by introducing unprecedented data about the amount of energy consumption of ads and posts, which will undoubtedly shape the addition of future legislations on the subject, highlighting the importance of regulating the use of social networks.

8. SOCIO-ECONOMIC ENVIRONMENT

This section encloses the socio-economic environment of this work, starting with the detailed budget needed for this study. Sections 8.2 and 8.3 describe the economic situation of Facebook and Twitter because of online advertising. The last section, 8.4, focuses on the economic and environmental impact resulting from this work.

8.1 BUDGET OF THIS WORK

In this section, every resource, both human and non-human, needed to carry out this project are detailed in Table 2, along with the amount of money that is outlaid due to the acquisition of these resources.

Table 2. Budget of resources

RESOURCE	BUDGET
Microsoft Surface Laptop	900€
Google Colaboratory Pro+	51€
Cloud Services	70€
Stationery	50€
Project Manager	4,320€ (60€/hour)
Programmer	12,690€ (30€/hour)
Maintenance	875€ (25€/hour)
TOTAL	18,956€

It is worth mentioning that the budget detailed above for every human resource in this work was calculated as a result of the 530 hours employed in the evolution of this work and detailed in section 9.

Adding the budget of every resource specified in Table 2, amounts to a total of 18,956€ for the development of this work.

8.2 FACEBOOK'S REVENUE

Facebook is estimated to generate around 97.5% of its total revenue through advertising, making it its indisputable primary source of revenue [29]. The amount of money made per year is currently changing at an increasing rate and showing no signs of slowing down, as the number of users is also increasing.

As it is clearly shown in Figure 9, there is clear yearly progression in the amount of money earned by Facebook ads in the United States over the years. In 2018, Facebook generated around 25 billion USD and in 2023, it raised more than 50 billion USD, doubling its ad income.

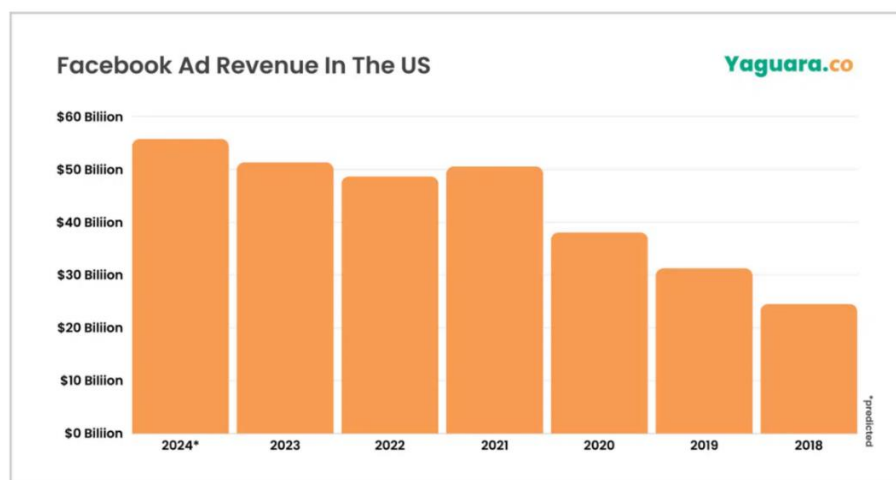


Figure 9 – Facebook ad revenue evolution in the U.S. Source: Yaguara.co

However, an exceptional increase in revenue occurred in 2021, as shown in Figure 9, maybe due to the COVID-19 outbreak and an unexpected increase in social media use, which was then corrected in 2022 after most social restrictions ended.

Considering that there are only about 240 million Facebook users in the U.S. [30], from a total of around 2.24 billion users worldwide, the yearly revenue obtained by Mark Zuckerberg's company from online advertising is simply outrageous. This amount of money does not come without cost, as the amount of energy being consumed from their ads, as explained in section 6, is as appalling as the money they are making.

Furthermore, the estimated amount of revenue generated by Facebook ads in the U.S. in 2024 is around 55 billion USD, about an 8% increase since the previous year.

8.3 TWITTER'S REVENUE

In 2022, Elon Musk completed the purchase of Twitter [31], and since then there have been many changes within the social network that affected their revenue streams significantly. Not only did the name get rebranded to X, but the social network got privatized and shares could no longer be bought [32].

In addition, Musk's controversial decisions during his time as CEO, such as firing thousands of staff members or limiting the number of posts users can see per day to 600 [32], led to huge losses in advertisement revenue, as the social network became less attractive to advertisers.

The combination of these factors severely affected the revenue that Twitter obtains from ads. In 2021, the social network achieved an all-time personal record in advertising revenue of about 4.5 million USD [33]. In 2022, the year in which Elon Musk started the acquisition of the company, the revenue decreased to 4 million USD, and in 2023, when Musk owned the whole company, the ad revenue drastically dropped to 2.5 million [33], their worst numbers since 2017.

Nevertheless, in 2023 Twitter obtained more than double their revenue in 2022 due to subscriptions and licensing, increasing from 0.4 million USD to 0.9 million USD. Still, the company's total revenue has been significantly reduced since then.

So, despite the recent losses Twitter has suffered, their highest source of revenue still comes from ads, which is an issue because the energy consumption due to advertising on Twitter is still very high, as their advertising reach encloses many users.

As a result, there is an urge in today's society to reduce energy consumption while Facebook and Twitter are both ignoring this fact and making huge amounts of money from this energy crisis.

8.4 IMPACT OF THIS WORK

As it has been presented before in this study, there are many users, and a lot of advertising that generates a high amount of revenue, but there is no record of an investigation into the amount of energy that this whole process consumes.

This work aims to give a realistic estimation of the amount of energy consumed due to the use of ads on social networks, and as seen in section 6, the situation is alarming.

Therefore, this work can have a significant impact on how social networks and the regulatory parties will act.

As explained in section 7, the EU is currently on a race to reduce the energy consumption in the region, and the data presented in this work gives a unique opportunity to take a huge step towards that goal of creating a more environmentally friendly society.

It is obvious that these social networks are making ridiculous amounts of money due to ads, but to the best of knowledge, they are taking no action to contribute to energy consumption. If they realize their influence on this issue and agree to collaborate with the corresponding parties, the energy consumption on social networks could be slowed down while maintaining a large amount of revenue from ads.

Either way, some sort of action is needed because the number of social media users is increasing and with it, the energy consumption, so this work can be a starting point for social media platforms and regulatory bodies to come together and elaborate the most adequate regulations to contribute to reducing the carbon footprint.

Another party that is involved in this narrative are advertisers. If the data obtained from this work can reach them, they would have the power to choose in which social media platforms to publish their ads based on how green they are, and therefore contribute to a greener society. If a large number of advertisers make their small contribution, the overall effect would be huge, and the environmental impact of ads on social media would be significantly decreased over time.

Lastly, the remaining party in this scheme are the social media users. Studies like this one allow these users to become aware of this issue and make informed decisions on which platforms to use and which ones to avoid.

If the loading time and energy consumption of Facebook ads were to be reduced by only 1% and considering that there are 2.11 billion daily active users [21] and every user visualized 70 ads per week [22], the yearly energy consumption would be decreased by 614,432 kWh.

Considering that for the second half of 2023 the average electricity price for household customers in the EU was 0.2847€/kWh [34], and the reduction mentioned above, about 174,928€ would be saved in energetic cost for the end users on a yearly basis.

9. PLANNING

As in any other project, a concise planning of the phases of this study should be designed to ensure a better execution and a more successful distribution of work.

To this end, a Gantt diagram was created using an open-source tool [35], which describes the different duration of each of the main phases and tasks to accomplish in this work, as well as a visual representation of the process to follow over the months in which this project takes place, shown in Figure 10.

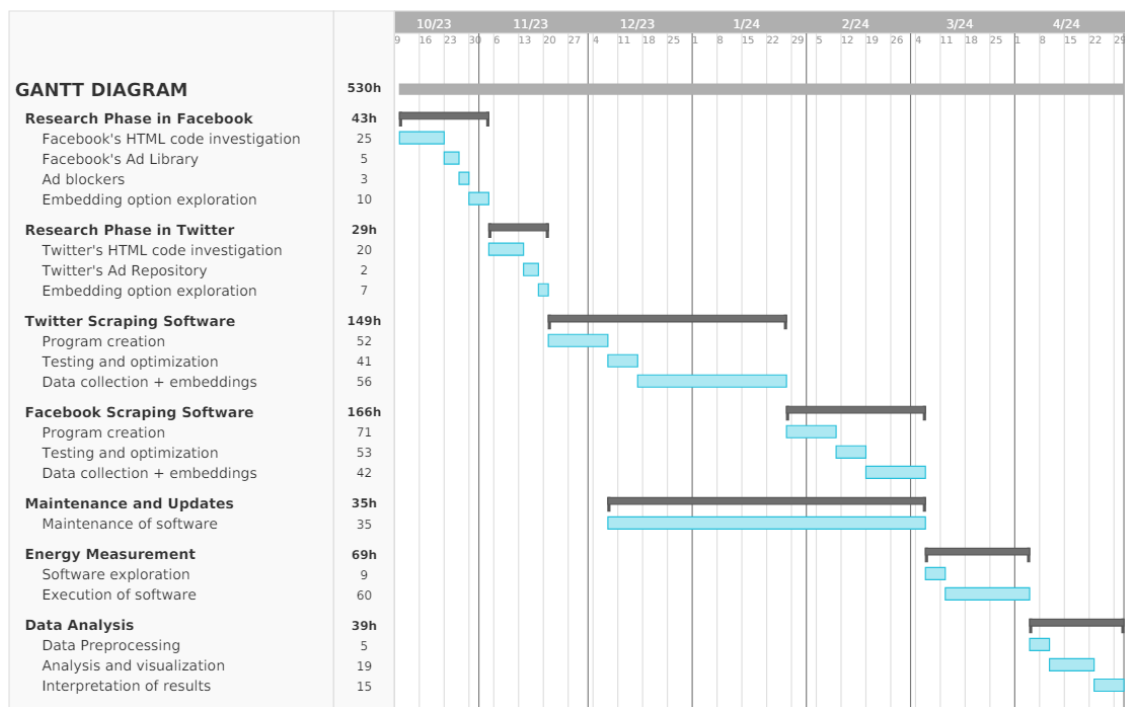


Figure 10 – Gantt diagram of the project

As Figure 10 shows, the development of the whole project implicated 530 hours of constant work, except for weekends. The duration of the project lasts from the 10th of October 2023 to the 30th of April 2024.

10. CONCLUSIONS AND FUTURE OF WORK

To conclude with, the goals and expectations previously set by this work will be analyzed in section 10.1, and some suggestions on how to contribute to this work or continue working on this subject will be detailed in section 10.2. In section 10.3, the conclusions of this work will be described.

10.1 OBJECTIVES MET

To begin with, the objective of finding a way of identifying all elements of an ad in both Facebook and Twitter was successfully completed, and it has a lot of value, considering that there is no free open-source API or software that can do this. Also, the software explained in this work was optimized in terms of capacity of ads that can be obtained per execution, and in time consumption, as the stop timers throughout the code were tested repeatedly.

Moreover, as there were many similarities between ads and posts in both social networks, it was very straightforward to adapt the software for ads to also obtain posts, which increases the value of this work. This also led to a very interesting comparison between ads and posts within Facebook and Twitter, which clearly determined that ads are significantly more energy consuming than posts.

The goal of measuring the energy consumption of ads was also successfully met, as a decent estimation of the amount of energy in kWh was obtained in both social networks by using the data from significant datasets.

10.2 FUTURE LINES OF WORK

As a result of this work, there are many future lines of work that can be explored. One of them is carrying out the same process for other social networks with both similar and different characteristics. For example, LinkedIn also provides an embedding option like Facebook and Twitter, so the software could be adapted to get LinkedIn ads and posts too.

On the other hand, a social network like TikTok, which only has ads and posts with videos, could be interesting to explore and find out the energy consumption it is

generating. It is likely to be higher than other social networks as it was found in this work that video ads and posts tend to consume more energy.

Also, an addition to this work could be having a larger dataset, especially for ads, as the limitations imposed by both social networks make it hard to obtain a very large dataset over a short period of time. Nevertheless, as indicated by researchers, ads only represent about 10-15% of the total information received by users on their Facebook feed [22], so it is completely normal that there are less ads than posts in the dataset. Still, a larger dataset would lead to more robust results and a more accurate estimation of the energy consumption per ad.

Another work in process could be carrying out the energy measurement phase with different electronic devices, as this work was solely executed using a Windows laptop. By using different devices, more accurate results and a better estimation of the energy consumption would be obtained.

10.3 CONCLUSIONS

During this study, a total number of 7,277 ads and posts have been retrieved from Facebook and Twitter, along with several parameters, such as number of likes or shares. The most relevant conclusions obtained from this dataset include the following:

1. Correlation factors
 - All ads and posts from both Facebook and Twitter are influenced by their loading time.
 - Facebook ads are also strongly influenced by the number of internal calls and by their visual content, whether it contains images or video. Facebook posts also have this influence, but not as strong as in ads
 - Twitter ads are slightly influenced by their popularity, while posts do not show any sign of influence.
2. Energy consumption
 - Ads on Facebook consume 1.6 times more energy and loading time than posts.
 - Ads on Twitter consume 1.75 times more energy and take 1.5 more loading time than posts.

3. Facebook vs Twitter

- Facebook consumes around 8.3 times more energy than Twitter in posts and 8.7 times more in ads.
- Facebook takes 6.2 times more than the duration it takes for Twitter to load posts and 6.7 times more for ads.
- Facebook consumes approximately 1.2 times more energy than Twitter in ads and 1.4 times more in posts per millisecond of loading time.

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